COUPLING TRAJECTORIES AND PARTITION STABILITY IN LATTICE YANG–MILLS THEORY

Abstract. We construct a coupling trajectory and its common vacuum normalization from admitted small-field formation, coefficient and large-field closure estimates. Uniform block drift gives a protected terminal window, arbitrarily fine first-passage cutoffs and the leading logarithmic ultraviolet law without positivity of each increment. Frozen-choice analytic estimates give tuned bare couplings under their stated prefix domains. Component bounds and label counting control the normalized zero-source history sum; a retained small-field event supplies the matching partition lower bound. The local analytic constructions, the one-loop coefficient and the quantitative history estimates are explicit inputs. No complex-source stability or continuum limit is asserted here.

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1. Introduction

Renormalization produces a new action coefficient before it has finished restoring the form of the effective measure. To iterate the construction, that computed coefficient must first be admitted on the domains required by the remaining operations. A proof that assumes the next complete effective density while checking this admission is circular. We give an induction in which the current history data determine the regular output and its coefficient, the coefficient extends the numerical frame, and the large-field operation then restores the next hypothesis. The common vacuum normalization is completed with the remainder returned by that last operation.

Two further issues are separate from this induction. A positive average drift need not make every coupling increment positive; first passage must allow bounded backward excursions. Also, a stochastic realization preserves total mass without bounding it. The partition estimates require quantitative bounds on actual components, the count of their labels, the retained integrals, and a lower-bound event with the same vacuum normalization.

The regular-iteration companion [1] supplies bounds along prescribed admissible prefixes. Here the one-step coefficient and closure inputs are used to produce a prefix to which those bounds apply. The scalar forward-profile estimate in that companion also has a backward fixed-point application. We keep this homogeneous profile argument distinct from tuning the full history-dependent trajectory, whose marked terms have separate analytic domains.

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Bałaban's effective-density estimates and stability bounds provide the conditional renormalization framework [2, Theorem 2 and Corollary 3, pp. 263–264] and [3, Theorem 1 and (0.1), pp. 355–356]. The running-coupling hypothesis is retained explicitly in the latter theorem. Those results do not by themselves supply the common trajectory or the coefficient identifications required here. We use actual local formation, geometry, normalization and history estimates as named hypotheses, and prove the ensuing induction, arithmetic, analytic tuning and partition deductions.

The partition bounds are uniform in cutoff and compatible finite volume at a fixed sufficiently small terminal coupling. The trajectory and tuning estimates state their dependence on the inverse-coupling floor. Complex-source estimates, transfer to infinite-volume Gibbs states and limiting field constructions require further results. The present arguments do not establish those conclusions.

2. Finite measures and the common vacuum normalization

Fix an odd integer $L \geq 3$, a length $\ell > 0$, and a finite cutoff index $K' \in \mathbb{Z}_{\geq 0}$. The physical torus is $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^4$. At level j its lattice Λ_j has mesh $L^{j-K'}$, for $0 \leq j \leq K'$. We require $\ell L^{K'-j}$ to be a positive integer and require all partitions used by the construction to be compatible with these periods. Write

$$a_{K'} = L^{-K'}, \qquad N_j = |\Lambda_j| = \ell^4 L^{4(K'-j)}.$$

The geometric admission conditions below may require a larger period than this divisibility condition alone.

A configuration assigns $U_b \in \mathrm{SU}(2)$ to each positive bond, with $U_{\bar{b}} = U_b^{-1}$ on the reverse bond. Every unoriented plaquette is counted once, with one fixed orientation. Set

$$A(U) = \sum_p \left(1 - \tfrac{1}{2} \operatorname{Tr} U_p\right), \qquad d\mu_{0,\beta}(U) = e^{-\beta A(U)} \prod_b dU_b, \qquad Z_{K',\ell,\beta} = \mu_{0,\beta}(1), \tag{2.i}$$

where dU_b is Haar probability measure and $\beta > 0$. The trace of an SU(2) matrix is real. On complexified link charts we use this same trace polynomial as the holomorphic extension of the action. The finite partition function is positive: the integrand is a strictly positive continuous function on a compact product. The initial presentation has this Wilson action, one empty history, and no effective regular, remainder or boundary families. Its membership in the admitted initial class also requires the geometric and bare-boundary inputs stated with that class.

At a later level the state includes the current bond configuration, the retained fields, and the discrete history records. We use the Borel sigma field on each finite-dimensional field space and the disjoint-union sigma field for the records. A stochastic kernel is a measurable probability kernel on these spaces. For a finite positive measure μ and a kernel Q, the measure μQ is defined by $(\mu Q)(F) = \int Q(F)(u)\, d\mu(u)$ for bounded Borel F. Each fixed real operation used here is supplied as such a kernel or as a composition of the operations in the next lemma. An analytic expansion of its density may have summands of either sign; that does not change the underlying positive measure.

Lemma 2.1 (Mass and retained variables)**.** Suppose a finite positive measure is transformed by finitely many of the following operations: a Borel test with finitely or countably many outcomes that partition the entire state space; a draw from a stochastic kernel; or a measurable map, including a map which forgets variables. Sum over all test outcomes. The total mass is unchanged.

More specifically, let m be a finite positive measure on $\mathsf{A} \times \mathsf{B}$ and let κ be a stochastic kernel from A to C. Adjoin a draw from κ and then retain only the C coordinate. The result is

$$\big[m(\kappa \otimes 1)\big]_{\mathsf{C}} = m_{\mathsf{A}}\kappa. \tag{2.ii}$$

These statements hold for a fixed signed or complex measure if all displayed integrals are absolutely integrable. ∎

Proof. For a test with indicators χ_r, the sum of the masses is $\int \sum_r \chi_r \, d\mu = \mu(1)$ because $\sum_r \chi_r = 1$. For a draw, $Q1 = 1$ gives $(\mu Q)(1) = \mu(1)$. For a map f, $f_*\mu(1) = \mu(1)$. Composition proves the first assertion. To prove eq. 2.ii, test both sides on a bounded Borel function F on C. The left side is

$$\int_{\mathsf{A}\times\mathsf{B}} \left(\int_{\mathsf{C}} F(c)\, \kappa(a, dc) \right) m(da, db).$$

The inner integral depends only on a, so integration against the marginal m_{A} gives the right side. Fubini's theorem applies because the measure is finite and F is bounded. Absolute integrability gives the same argument for signed or complex measures. □

For actual Borel block averages M_j between the successive link spaces, write $T_j = (M_j)_*$. Their bare pushforwards satisfy

$$T_{n-1} \cdots T_0 = (M_{n-1} \circ \cdots \circ M_0)_*.$$

Indeed both sides integrate a bounded Borel function against its composition with $M_{n-1} \circ \cdots \circ M_0$. For $n = 0$ both are the identity. This concerns the bare averaging maps; the full real operations also keep the stipulated history variables.

Undefined local means, logarithms or minimizers are assigned a failed-test outcome, so they do not leave an unassigned part of the configuration space. A gauge or frame draw precedes every test that depends on that draw, and its value is retained in the history until it is forgotten by a specified map. Replacing such a draw by multiplication by a gauge-fixing weight is valid only under the separate density identity that justifies that replacement. In particular, the mass passing a subsequent frame-dependent test must be averaged with the frame actually used by that test.

For a fixed real run let Q_j be its complete operation on the extended states at step $j \to j+1$, and put $\mu_{j+1} = \mu_j Q_j$. Its stochastic realization is part of the actual local construction. If reference laws are fixed in advance, the same Q_j acts linearly on a family of input measures. An operation whose reference law is recomputed from each density cannot be declared linear for this reason. At zero source, lemma 2.1 gives $\mu_j(1) = Z_{K',\ell,\beta}$. This identity supplies no bound uniform in the cutoff after a vacuum normalization: every multiple $c\nu$ of a probability measure also satisfies the same mass-preservation identity, for any $c > 0$.

Lemma 2.2 (Bounded insertions and positive history kernels)**.** Let μ be a finite positive measure with positive mass on a standard Borel bare state space X. Let K_h be positive measurable kernels from X to standard Borel retained state spaces Y_h, indexed by a finite or countable set of records, with $\sum_h K_h(u, Y_h) = 1$ for every u. For a bounded Borel function $F : X \to \mathbb{C}$, put

$$\tau_h(B) = \int K_h(u, B)\, d\mu(u), \qquad \tau_h^F(B) = \int F(u) K_h(u, B)\, d\mu(u).$$

Then $|\tau_h^F| \le \|F\|_\infty \tau_h$. The density $F_h = d\tau_h^F / d\tau_h$ is a version of the conditional expectation of F given the record and retained state, under the probability law on the disjoint union of the spaces $X \times Y_h$ whose record-h part is $\mu(du)K_h(u, dy)/\mu(1)$. In particular,

$$\frac{\int F\, d\mu}{\mu(1)} = \frac{\sum_h \int F_h\, d\tau_h}{\sum_h \tau_h(1)}, \qquad |F_h| \le \|F\|_\infty \quad \tau_h\text{-a.e.} \tag{2.iii}$$

If f_h is Borel and $F(u) = f_h(y)$ almost everywhere in the joint law of record h, then $F_h = f_h$ almost everywhere on Y_h. If a measurable symmetry r with measurable inverse preserves μ, permutes the records and transports their spaces and kernels by $K_{rh}(ru, rB) = K_h(u, B)$, then $r_*\tau_h = \tau_{rh}$ and $r_*\tau_h^F = \tau_{rh}^{F \circ r^{-1}}$. ■

Proof. For each measurable partition of B, the sum of the moduli of the inserted integrals is at most $\int |F(u)| K_h(u, B)\, d\mu(u)$. The variation bound follows. The Radon–Nikodym theorem gives F_h and its essential bound. Testing the defining identity against functions of (h, y) identifies the conditional expectation. It also gives the asserted version when F is already determined by the

retained state. The sums are absolutely convergent by the variation bound and $\sum_h K_h(u, Y_h) = 1$. Summing the inserted and uninserted masses gives eq. 2.iii. For the covariance assertions, change u to ru in the measures of rB for record rh. □

For the Wilson measure one may take $F(u)$ to be a bounded gauge-invariant function of the iterated bare average. Its conditional value is the corresponding function of the retained fields when those fields determine it. The complete history operation may replace other fields, so the insertion formula uses the joint law of the original field and the retained state.

We now fix the numerical subtraction used throughout the paper. The new regular family at level $j+1$ is $\mathcal{E}_{j+1}(X, P, z)$, and the new remainder family is $\mathcal{R}_{j+1}(X, P)$. Their carrier X is a specified finite union of localization cells, z is a pointing index, and $P = (V, J)$ is the admitted field-current pair. The trivial pair is $P_\circ = (\mathbf{1}, 0)$. Values at this pair use the actual localization and its fixed logarithm branches. The sums below are finite or absolutely convergent in the supplied rooted norms. They range over the complete periodic reference families, with the same convention at every history.

Let $\mathfrak{n}_j$ be the field-independent numerical factor assigned to the actual step. Define

$$E_{(0)} = 0, \qquad E_{(j+1)} = E_{(j)} + \mathsf{e}_j,$$
$$\mathsf{e}_j = \mathfrak{n}_j + \sum_X \left\{ \sum_z \mathcal{E}_{j+1}(X, P_\circ, z) + \mathcal{R}_{j+1}(X, P_\circ) \right\}. \tag{2.iv}$$

The number $E_{(j)}$ may depend on the run, cutoff and torus. It is common to all histories on that fixed torus. Missing local vacuum contributions in a particular history remain in its normalized activity; they are not subtracted a second time. The new remainder in eq. 2.iv is the one returned by the completed large-field operation. Thus this recursion is completed after coupling admission and closure, and is not an input to the calculation of that coupling.

At step j, the positive number x_j is the action weight, and the positive number θ_j is its frozen reference weight. For $\tau_j = x_j/\theta_j$, one assigned Gaussian factor in a resolved SU(2) fibre is $-\frac{3}{2}|\mathfrak{X}_j| \log \tau_j$, where $\mathfrak{X}_j$ is its set of free non-pivot bonds. Each such bond has three real coordinates. The logarithm is based at $\tau_j = 1$ on the admitted connected relative disc. Haar, gauge and reference-normalizer constants are assigned by the same actual formula. We use the Gaussian family normalized at the trivial datum, as in [1, §4]; its removed reference values are included once in $\mathfrak{n}_j$. A presentation retaining the original Gaussian vacuum values is equivalent only after the opposite change to $\mathfrak{n}_j$, as proved below. The gauge-compensation factor in an interacting cut remains part of that integral; its unrestricted Gaussian integral cannot replace it there.

Lemma 2.3 (Common normalization and redistribution of constants)**.** Let the actual finite measures and the real numbers in eq. 2.iv be supplied as above. Put

$$\nu_j = e^{-E_{(j)}} \mu_j, \qquad \widehat{Z}_j = e^{-E_{(j)}} Z_{K',\ell,\beta}.$$

Then

$$\nu_{j+1} = e^{-\mathsf{e}_j} \nu_j Q_j, \qquad \nu_j(1) = \widehat{Z}_j. \tag{2.v}$$

Replacing a local family by that family minus a field-independent number, and adding the sum of the removed numbers to $\mathfrak{n}_j$, leaves e_j, $E_{(j)}$ and these normalized measures unchanged. The same assertion holds if the numbers are moved in the opposite direction. ■

Proof. Since Q_j is fixed for this run, multiplication of its input measure by the scalar $e^{-E_{(j)}}$ commutes with its action. The identity $E_{(j+1)} = E_{(j)} + \mathsf{e}_j$ therefore gives the first equation. Taking total mass and using lemma 2.1 gives the second. In eq. 2.iv, subtracting constants from the vacuum values cancels their sum added to $\mathfrak{n}_j$. Absolute convergence permits that redistribution. Induction on j leaves every accumulated number unchanged. □

Redistribution is made in the complete density. If a history omits a carrier, its residual numerical factor changes by the corresponding omitted constant, so the physical measure stays fixed.

The coupling subtraction has a different role. The coefficient b_{j+1} is read from the newly formed complete regular family, using the normalized classical insertion whose coefficient is one. Subtracting b_{j+1} times that insertion changes the classical exponent from $-x_jA$ to $-(x_j - b_{j+1})A$. Its vacuum value is zero, since $A(\mathbf{1}) = 0$. The new young remainder is retained in its designated class; its vacuum value contributes to eq. 2.iv after closure. One does not add its quadratic coefficient as a second coupling subtraction. Older admitted remainders already enter the computed full regular coefficient through the actual regular step.

The counterterm bookkeeping also applies to a spatially varying record of completed steps. Put $A(w, U) = \sum_p w(p)(1-\frac{1}{2}\operatorname{Tr} U_p)$ for a plaquette weight w. If the actual history assigns a counterterm $b_i\phi_i(p)$ at plaquette p, set $w_0(p) = \beta$ and $w_i(p) = w_{i-1}(p) - b_i\phi_i(p)$. At a fixed read of the field U, linearity in the weights gives

$$-A(w_k, U) - \sum_{i=1}^{k} b_i A(\phi_i, U) = -\beta A(U).$$

The profile ϕ_i records where step i was performed. On the all-small-field history it is one for the corresponding complete counterterm. The displayed identity concerns the actual assignment of coefficients and does not change the background or replace its tests.

For the real history estimate, the actual profiles on this same plaquette field are required to satisfy $1 \geq \phi_1(p) \geq \cdots \geq \phi_k(p) \geq 0$. If $x_j > 0$ and $b_j = x_{j-1} - x_j$, summation by parts gives

$$w_k(p) = x_0(1 - \phi_1(p)) + \sum_{j=1}^{k-1} x_j(\phi_j(p) - \phi_{j+1}(p)) + x_k\phi_k(p) \geq 0. \tag{2.vi}$$

The coefficients are nonnegative and sum to one. This proves positivity without requiring $b_j \geq 0$. For $k = 0$ use $w_0 = x_0$. The native geometric condition gives this ordering when each profile is supported in its action region and equals one on its one-cell interior, and the next action region lies in that interior. Wherever the next profile is positive, the preceding one is then one. On a stock where the actual profiles equal one for $j < t$ and zero for $j \geq t$, the weight is x_{t-1}. That stock identity is a geometric input about the actual action profiles. All these action terms use the same field read. An inner retained island uses its own assigned field and Wilson factor. Every allocation of energy shares must consume at most the original nonnegative weight on each plaquette of its own field, leaving a nonnegative residual.

All constants in the later estimates are uniform in the cutoff, the length of an admitted prefix and the compatible torus, except where a dependence on the terminal coupling or volume is displayed. Geometric constants, localization parameters and analytic reserves are fixed before the final inverse-coupling floor. A numerical inequality about the resulting increments does not by itself provide the admitted local maps or their common domains.

2.1. **The logarithmic cost of a local normalizer.** The numerical factors in a history must be charged on their actual carriers. One elementary estimate explains the logarithmic cost of the scalar Haar normalizers used in that assignment.

Lemma 2.4 (A truncated Haar normalizer)**.** For $y \geq 1$, let $\chi_y : \mathrm{SU}(2) \to [0, 1]$ be Borel. Suppose a fixed $a_* > 0$ satisfies $\chi_y(U) = 1$ whenever $\|U - I\|_{\mathrm{op}} < a_* y^{-1/2}$. With Haar probability measure define

$$z(y) = \int \chi_y(U) e^{-y(1-\frac{1}{2}\operatorname{Tr} U)}\, dU.$$

There are constants $0 < c \leq C < \infty$, depending only on a_*, such that

$$cy^{-3/2} \leq z(y) \leq Cy^{-3/2}, \qquad |\log z(y)| \leq C + \tfrac{3}{2}\log y. \qquad \blacksquare$$

Proof. The class angle $\phi \in [0,\pi]$ has Haar density $(2/\pi)\sin^2\phi\, d\phi$. Also $1-\frac{1}{2}\operatorname{Tr} U = 1-\cos\phi$ and $\|U-I\|_{\mathrm{op}} = 2\sin(\phi/2)$. The inequalities $\sin\phi \le \phi$ and $1-\cos\phi \ge 2\phi^2/\pi^2$ on this interval give

$$z(y) \le \frac{2}{\pi}\int_0^\infty \phi^2 e^{-2y\phi^2/\pi^2}\, d\phi = Cy^{-3/2}.$$

Put $a = \min\{a_*/2, 1\}$. For $0 \le \phi < ay^{-1/2}$, the cutoff is one, $\sin\phi \ge 2\phi/\pi$, and $y(1-\cos\phi) \le a^2/2$. Integrating on this interval gives $z(y) \ge (8a^3/(3\pi^3))e^{-a^2/2}y^{-3/2}$. Finally $z(y) \le 1$ and this lower bound give the logarithmic estimate. □

For example, a fixed numerical factor $\sigma_0 > 0$ contributes $\log\sigma_0 - \frac{3}{2}\log x_j$ per assigned free bond, and a factor $z(x_j)^{-(L^4-1)}$ contributes $-(L^4-1)\log z(x_j)$ per assigned coarse cell. Each costs at most $C(1+\log x_j)$ at its own level. Their actual multiplicities and carriers come from the finite density formula. When a history omits such a carrier, its cost is charged there; it is not bounded by the number of final coarse sites without the geometric and weighted-integral estimates proved below.

The regional formula can be seen explicitly in the self-calibrated convention $x_{j-1} = \theta_{j-1}$. At a step with regions $\Lambda \subset \Omega$, write $B^{\mathrm{free}}_{j-1}(Y)$ for the number of incident free fine bonds in Y after pivot removal, and $\#_j(Y)$ for the level-j site count. Let σ_0 be the Haar density at the identity in the chosen exponential coordinates, let z_{j-1} be the scalar site-gauge normalizer. On $\mathfrak{g} = \mathfrak{su}(2)$, with its chosen Euclidean inner product and Lebesgue measure, set $\nu_{\mathfrak{g}} = \log\int_{\mathfrak{g}} e^{-|Y|^2/2}\, dY$. In the convention that keeps the unnormalized Gaussian local vacuum values in the regular family, the three elementary scalar contributions at this step are

$$\begin{aligned}(\log\sigma_0 - \tfrac{3}{2}\log x_{j-1})B^{\mathrm{free}}_{j-1}(\Omega) - (L^4-1)\#_j(\Omega)\log z_{j-1}\\ + \nu_{\mathfrak{g}} B^{\mathrm{free}}_{j-1}(\Lambda).\end{aligned} \tag{2.vii}$$

The vacuum values of the regular and remainder families on their actual ranges are added to this number. If a local Gaussian family is normalized at the trivial pair, its removed vacuum values are instead added once to the assigned number, as in eq. 2.iv.

The first term of eq. 2.vii comes from extracting σ_0 from each free-link Haar density and scaling its three real coordinates by $x_{j-1}^{-1/2}$. The second counts the L^4-1 noncentral site gauges per level-j block. The last is the scalar for expressing the fixed reference Gaussian on the three real coordinates of each counted bond as a probability measure. This normalization is an identity between reference densities before the cut is imposed. The cut and every gauge-compensation factor stay inside the resulting integral.

The last factor normalizes the product standard reference density. Any other field-independent covariance or pivot number keeps its prescribed allocation in the raw local Gaussian total or assigned scalar, with its trivial value counted once. It is not an additional evaluation of an interacting cut integral.

Proposition 2.5 (Regional counts and comparison of prefix scalars)**.** At a step $j-1 \to j$, let $Y \subset \Lambda_j$ and identify it with the union of its fine blocks. Count a positive bond in the regional normalizer when at least one endpoint is in Y, and remove one distinct fine pivot for each incident coarse bond, using the actual incidence convention. Use L^4-1 normalized site gauges per coarse site in Y. Let $d_{j-1}(Y)$ be the number of remaining incident fine bonds minus the number of these site gauges. Let $B^{\partial}_j(Y)$ be the number of coarse bonds with exactly one endpoint in Y. Then

$$\begin{aligned} d_{j-1}(\Lambda_j) &= 3(L^4-1)N_j,\\ d_{j-1}(\Lambda_j) - d_{j-1}(Y) &= 3(L^4-1)|Y^c| - \tfrac{1}{2}(L^3-1)B^{\partial}_j(Y).\end{aligned} \tag{2.viii}$$

The second line lies between zero and $3(L^4-1)|Y^c|$.

Suppose the actual normalizers assign the Gaussian scale $-\frac{3}{2}\log\theta_{j-1}$ to each free bond and the reciprocal Haar scale $+\frac{3}{2}\log\theta_{j-1}$ to each such gauge. Keep the remaining scalar prices, including the bounded ratios x_{j-1}/θ_{j-1}, in the prescribed step constants. The logarithmic part left after subtracting the full-torus value is $\frac{3}{2}\log\theta_{j-1}$ times the second line of eq. 2.viii. In particular it is a nonnegative cost when $\theta_{j-1}\geq 1$.

If the remaining scalar price on a full level is bounded in absolute value by CN_j, and a fixed $C_{\mathrm{exc}}\geq 0$ satisfies $\theta_i-\theta_j\geq -C_{\mathrm{exc}}$ for $i\leq j$, then on one fixed run, for $0\leq k\leq n\leq K'$,

$$E_{(n)}-E_{(k)}=-\tfrac{9}{2}(L^4-1)\sum_{j=k}^{n-1}(\log\theta_j)N_{j+1}+O(N_k). \tag{2.ix}$$

Consequently replacing the prefix normalization $E_{(k)}$ by $E_{(n)}$ changes the normalized mass by a factor between e^{-CN_k} and $e^{C(1+\log\theta_k)N_k}$, for $\theta_j\geq 1$. ■

Proof. There are $4N_j$ positive coarse bonds and $4L^4N_j$ positive fine bonds. The full free-bond count is $4(L^4-1)N_j$. The full site-gauge count is $(L^4-1)N_j$, proving the first line. Count coarse bond endpoints in Y^c to get $4|Y^c|-B_j^{\partial}(Y)/2$ bonds with both endpoints outside Y. There are L^3 fine crossings per coarse face, so the analogous fine count is $4L^4|Y^c|-L^3B_j^{\partial}(Y)/2$. Each omitted pivot corresponds to a coarse bond with both endpoints outside Y. Subtract the omitted pivot and site-gauge counts from the omitted fine count to get the second line. The bound $B_j^{\partial}(Y)\leq 8|Y^c|$ gives its nonnegativity for $L\geq 3$. Its upper bound follows by dropping the boundary term. The count applies on the periodic lattice, counting oriented occurrences if projected edges have the same endpoints.

The signed scale price is $-\frac{3}{2}\log\theta_{j-1}$ times the difference of free-bond and site-gauge counts. Subtract its value on the full torus. This gives the sign and magnitude claimed for the regional omission. Summing the full-torus values in the once-counted recursion gives the principal term of eq. 2.ix. The remaining costs sum to $O(N_k)$ because $N_{j+1}=L^{-4(j+1-k)}N_k$. Finally $0\leq\log\theta_j\leq\log(\theta_k+C_{\mathrm{exc}})$ for $j\geq k$. Exponentiating the comparison with the sign in the normalized mass proves both factor bounds. □

3. Joint construction of the weights and presentations

Use the finite measures and the vacuum convention of section 2. To distinguish a chosen finite lattice from the stopping time, write K' for the finite cutoff and use K for first passage. At level j the period in its own units is $P_j=\ell L^{K'-j}$. Prefixes on different tori or cutoffs are compared through the prescribed common lifted reads.

3.1. **The local inputs.** For a real reference frame $\theta=(\theta_0,\dots,\theta_k)$, an admitted presentation through k records the following data. It records the weights x_j, their relative domains in the frame, the complete regular families of both chains and their common companions, and their coefficients at each created level. It also records the unpointed remainders with their birth cores, the boundary terms with their determining data, and the positive history measures with retained fields, tests, and local exponents. The exponents have the Wilson weight and the once-counted common scalar of eq. 2.iv. Each regular term in the action is vacuum subtracted and carries its own classical quadratic counterterm. Each remainder is vacuum subtracted.

The analytic class includes the declared real gauge covariance. History kernels are covariant under lattice isometries preserving all their partitions. Each localized regular or remainder family has translation covariance under the translations preserving its own partition. Full lattice-isometry covariance is required of the unrestricted pointed totals on jointly admitted real background pairs; it is not imposed on every filed local term. Identifications of complete reads transport the actual local formulas, including the support tests, retained variables, measure factors and branches. The assigned remainder and boundary spaces, together with the birth domains of standing operations,

remain part of the data. A bound on an activity does not identify its formula, or change the space in which a later step may read it.

We use three local inputs. They are uniform for the prescribed constants and for every admitted old prefix, with no hypothesis on a future trajectory.

(1) The initial Wilson presentation is admitted. At level k, the actual regular and boundary formation reads the given presentation and the frame through k. It supplies the new complete regular output, its absolute-radius unmarked counterpart, their common analytic restrictions and companions, and the boundary data required for the succeeding operations. The full regular list uses the old remainders that have passed their age and domain tests. The unmarked list has none. These formations use the same frozen reference, real fluctuation cut and physical coordinate choices on their common reads. They are available before a new frame entry is chosen. In particular the physical creation radii and covariance threshold are specified by the old reference θ_k.

(2) The physical coefficient is linear on the common creation companion. Each finite realization assigns that same companion read as its counterterm; wrapping terms remain in its finite action bounds. The unmarked and marked envelopes from the regular iteration apply on this prefix [1, Sections 3–6]. On the self-calibrated choice $\theta_j = x_j$ for $j \le k$, the actual counterterm has the form

$$b_{k+1} = b^0_{k+1} + f_{k+1} + q_{k+1}, \qquad |f_{k+1}| \le \epsilon_{\mathrm{u}}(x_k), \quad |q_{k+1}| \le \epsilon_{\mathrm{m}}(x_k). \tag{3.i}$$

The nonnegative envelopes tend to zero at infinity. The actual one-loop data are independent of the initial weight, admissible torus and unused finer horizon. We require

$$\left|\sum_{l=1}^{r}(b^0_l - c_0)\right| \le c_2 \quad (r \ge 0), \qquad c_0 = \frac{11N^2}{12\pi^2}\log L, \qquad N = 2. \tag{3.ii}$$

Here $c_2 \ge 0$. This is the determinant and trace input in the Wilson convention $x_0 = \beta = g_0^{-2}$. A Gaussian finite-dimensional integration identity alone does not compute c_0. For the quantitative estimate below the ordinary regular envelopes can be taken as

$$\epsilon_{\mathrm{u}}(y) \le C_{\mathrm{u}} y^{-1/2}(\log y)^{\mathfrak{p}_0}, \qquad \epsilon_{\mathrm{m}}(y) \le C_{\mathrm{m}} y^{-p_R}, \qquad p_R = (\kappa_0 - 4)/2,$$

on the supplied domains, with fixed $\mathfrak{p}_0 \ge 0$ and $\kappa_0 > 4$. Dropping the logarithmic improvement in the marked estimate only enlarges this envelope.

(3) For an admitted presentation and an admitted new weight, the actual large-field closure returns the successor presentation with the same constants and prolongs the old data. Admission requires the real tube, frame inequalities, geometric tests and coupling ceilings of the construction. Its conclusion supplies the next positive histories, regular and remainder bounds, boundary terms, and their covariance and common-read identities. At fixed torus and cutoff the actual measure transformation has the mass identity of lemma 2.1. The new final remainder and the step scalar for eq. 2.iv are supplied after this closure.

The common-read input includes the infinite companion and compatibility under unused finer cutoffs, with the same reference and weight prefix. For each bare weight above the fixed input floor, the Wilson presentation with $\theta_0 = x_0 = \beta$ is admitted on a cofinal family of sufficiently large tori and fine cutoffs. Given an admitted old prefix, its regular formation is available on such a family. If its computed next weight and chosen reference entry satisfy the numerical admission inequalities, assume there is a cofinal family on which the entire finite proposed extension has the geometric and analytic read admissions of the actual closure. This includes the contour, chart, determining-history and support conditions for the prescribed tests and outcomes. If two realizations contain the same complete lifted read with its margins, the prescriptions agree there, including earlier reads used in the new step. These are local inputs for the initial entry and for one computed extension at a time. Each realization uses its own finite measure and scalar constants.

Lemma 3.1 (A conditional construction of coherent coefficients)**.** Fix a real frame prefix and its localization parameters. Suppose the basic local coefficients on compatible tori are holomorphic on the specified complex read charts, with uniform bounds there. The charts include the actual determining collars. On each torus take the finite subgroup of translations preserving the admitted settings, charts, labels and rooted bounds. The coefficients have a specified sum rule whose total is covariant under this subgroup.

Suppose a nested cofinal sequence of finite covers eventually covers each fixed target torus and each bounded lifted read. The covers supply linear filing maps. A filing assigns lifted labels to their actual target labels and reads their entries by fixed chart pullbacks. For each fixed finite part of a filing these pullbacks take compact subsets of the target chart into compact subsets of the corresponding source charts. Assume the filings compose under coverings and restrictions. Each admitted target translation has compatible admitted lifts on sufficiently large covers, and filing commutes with these translations. The filed sums obey the actual cover identity for the prescribed total on each target. Fix $C_0, C_1 > 0$, $D_{\mathrm{cat}} \geq 1$ and $0 < q < D_{\mathrm{cat}}^{-1}$. On each fixed rooted chart assume there is a fixed catalogue of at most $C_0 D_{\mathrm{cat}}^m$ length-m labels, the same across all covers, with absent entries set to zero and entry bound $C_1 q^m$ and filing multiplicity at most $m+1$, uniformly over the covers. The target charts form a countable collection of finite-dimensional charts, with countably many discrete read types, labels and entries.

Then translation averaging preserves the local read, bounds and sum rule. Along one cofinal subsequence the averaged, filed entries have holomorphic limits on the common charts. The limits are translation covariant, compatible under the filing maps, and satisfy the prescribed sum rules. The compatible restrictions to bounded lifted reads give a subsequential infinite companion. For representatives in a box, the comb with a prescribed exterior root and entrance bond gives a local gauge rule. Transforming the input by a site gauge u transforms its representative by u outside the box and by the constant $u(o)$ inside, where o is the exterior root. ■

Proof. On a finite torus let T be the admitted translation subgroup, acting on formulas by translating the setting, the label and the read coordinates. For a coefficient family C use $|T|^{-1} \sum_{a \in T} a^* C$. If two settings have the same complete local read, their translates do also, so every summand has the same value on the two settings. This proves locality of the average. Convexity preserves the rooted bounds. A translation permutes the summands, proving covariance. Summing the averaged coefficients gives the average of the translated sum rules, which equals the original covariant total.

For a box representative let $g_U(x)$ be the ordered transport along the prescribed tree from o to x inside, and put $g_U = 1$ outside. With $U^g(x,y) = g(x)U(x,y)g(y)^{-1}$, each tree bond of U^{g_U} is the identity. The root value and the tree equations determine g_U uniquely. In the transformed field the tree transport is $g_{U^u}(x) = u(o)g_U(x)u(x)^{-1}$ inside. The product gauge $g_{U^u}u$ therefore equals $u(o)g_U$ inside and u outside. This is the stated transformation law and uses only the actual tree and its entrance read.

File the averages on the common charts. Uniform boundedness of each scalar entry gives a normal family on its finite-dimensional chart. Extract successively on compact exhaustions of the countably many charts and take a single diagonal subsequence. Each admissible target translation is realized by the stated lifts for all sufficiently large covers, so its identity passes to the local uniform limits. The compact containment for the pullbacks ensures local uniform convergence of every finite filed sum. For a rooted filing or sum rule the tail beyond length s is bounded uniformly by

$$C_0 C_1 \sum_{m > s} (m+1)(D_{\mathrm{cat}} q)^m,$$

which tends to zero. Pass first to the limit in each finite partial sum, then let s increase. Composability of the filing maps before the limit gives the corresponding identities between limits, and the supplied cover identity identifies the limiting sum with the prescribed total on its stated domain. □

The chart margins, compatibility of the filings and cover identity are inputs about the actual local operators. The limit supplies a coherent choice of basic coefficients. Exact agreement with an already fixed finite realization on a nonwrapping read still requires the stated common-read identification. Identification of the physical coefficient uses convergence of the actual responses on the common charts and the moment argument of [1, Proposition 6.3].

3.2. **An order for the constants.** In the local input order, the exponents and fixed separation fractions precede L and the initial linear cubes. The prescribed decay rates and their reserves come next, followed by the enlarged localization cubes and the final cell multiplier M. Fix the analytic reserves, norm caps and cutoff constants before the weak frame triple. The triple is fixed before the upper coupling ceiling γ is chosen. The bare-disc radius and the room for the admitted background continuations come after that ceiling, and before the final inverse-coupling floor and the physical choices of terminal coupling, compatible torus, bare weight and cutoff. Each local constraint is checked at its last free parameter after substitution of quantities determined by earlier choices. For example, a lower bound on a decay reserve must be met before imposing the larger cube size that pays for it, and the final coupling floor must allow the already fixed bare-disc and background rooms. The local analytic and geometric inequalities used at these stages are the supplied ones. The reserve for retained integrals is a separate choice. Fix it after the required loads and their charge constants, and before any coupling ceiling or inverse-coupling floor used to pay those charges.

The class conditions at a chosen parameter must be finitely many floors, ceilings, or endpoints of a nonempty window, with earlier data fixed. A uniform estimate indexed by a later observable exponent is used only at the finitely many exponents fixed at this stage.

Lemma 3.2 (Order of the numerical choices)**.** Suppose the local inputs admit the preceding finite order of choices. Fix c_0, c_2 from eq. 3.ii, and choose $C_\beta \geq c_0 + 2c_2 + 1$. Set

$$\begin{aligned} c_5 = 2c_2, \qquad \lambda_{\mathrm{w}} = c_0/4, \qquad B_{\mathrm{w}} = C_\beta + 1, \\ \lambda_{\mathrm{s}} = c_0/2 - \varsigma, \qquad B_{\mathrm{s}} = C_\beta + \varsigma, \qquad 0 < \varsigma \leq \min\{c_0/4, 1\}. \end{aligned} \tag{3.iii}$$

After fixing the weak triple $(\lambda_{\mathrm{w}}, c_5, B_{\mathrm{w}})$, one can impose all the following numerical conditions on a common floor Y:

$$\epsilon_{\mathrm{u}}(y) + \epsilon_{\mathrm{m}}(y) \leq \min\{c_0/4, 1\}, \qquad \epsilon_{\mathrm{m}}(y) \leq \varsigma \qquad (y \geq Y). \tag{3.iv}$$

It can also impose $Y > \max\{2c_5, \gamma^{-2} + c_5 + 1\}$, where γ is the supplied upper coupling ceiling. For the fixed cell exponent $r > 0$ it can impose

$$\left(\frac{\log(y + 2B_{\mathrm{s}})}{\log y}\right)^r < L \qquad (y \geq Y).$$

If the required reserve is $\varsigma = C_* \gamma^{\kappa_0 - 8}$ with $C_* > 0$ fixed before γ, then $\kappa_0 > 8$ permits that reserve in eq. 3.iii. ■

Proof. With earlier data fixed, finitely many floors on a free upward parameter are combined by their maximum. Finitely many positive ceilings are combined by their minimum. A window may be chosen only after its nonemptiness has been checked in the preceding parameters. Apply this rule through the supplied local choices. The weak triple is fixed before taking the constants of the marked bound. If the displayed reserve is used, its positive exponent allows the ceiling γ to be reduced until ς lies in the stated range. Both error envelopes tend to zero with the fixed earlier constants. This permits eq. 3.iv and the other finite floors to be met together. The strong triple satisfies $\lambda_{\mathrm{s}} \geq \lambda_{\mathrm{w}}$ and $B_{\mathrm{s}} \leq B_{\mathrm{w}}$, so it needs no new estimate on a different weak class. The logarithmic ratio in the cell condition tends to one, allowing its floor at the same stage. □

The source uses rounded cells based on a running minimum. Fix the power $r > 0$ chosen with the geometry, and write

$$R(y) = \min\{L^m : m \in \mathbb{Z}_{\geq 0},\ L^m \geq (\log y)^r\}, \qquad R_j = R\Big(\min_{0 \leq i \leq j} x_i\Big). \tag{3.v}$$

The real cutoff can be chosen on any cofinal class satisfying the actual geometric admission conditions. The numerical floor in the source is $P_j \geq 2L^{11} M R_j$ at each used level, where M is the fixed localization side in level units. If a construction needs a prospective floor at $j + 1$ before x_{j+1} is known, $P_{j+1} \geq 2L^{11} M R_j$ is sufficient for that numerical condition. The local contour and history-domain tests remain separate inputs.

3.3. **Admission and closure.**

Theorem A (The joint trajectory through first passage)**.** Assume the three local inputs and the cofinal admission stated in section 3.1 on weak prefixes, with the constants chosen as in lemma 3.2. On an admitted old prefix the weak frame inequalities are

$$\lambda_{\mathrm{w}}(j - i) - c_5 \leq x_i - x_j \leq B_{\mathrm{w}}(j - i) \qquad (i \leq j).$$

Fix $h \geq Y$ and $\beta \geq h + B_{\mathrm{s}}$. Start from the admitted Wilson presentation, put $x_0 = \beta$, and use $\theta_j = x_j$ as each new entry is admitted. The prescribed construction forms the real coefficients and actual presentations through

$$K = K(\beta, h) = \min\{j \geq 0 : x_j \leq h + B_{\mathrm{s}}\}. \tag{3.vi}$$

The coefficients and the stopping index are common to all admitted tori and unused finer horizons that realize this prefix. The joint vacuum recursion is eq. 2.iv with the same convention in each finite realization.

The stopping index is finite. For all $0 \le i \le j \le K$,

$$\tfrac{3}{4}c_0(j-i) - c_5 \le x_i - x_j \le B_{\mathrm{s}}(j-i), \qquad \lambda_{\mathrm{s}}(j-i) - c_5 \le x_i - x_j. \tag{3.vii}$$

The terminal value is protected by the preceding window:

$$h < x_K \le h + B_{\mathrm{s}}, \qquad \frac{\beta - h}{B_{\mathrm{s}}} - 1 \le K \le 1 + \frac{\beta - h - B_{\mathrm{s}} + c_5}{\lambda_{\mathrm{s}}}. \tag{3.viii}$$

In particular, for fixed h the stopping depths are unbounded as β tends to infinity. ■□

Proof. At $k = 0$ the frame inequalities have no nontrivial segment. If $x_0 = h + B_{\mathrm{s}}$, the initial presentation is already terminal. Otherwise suppose the construction has reached k without stopping. In particular $x_k > h + B_{\mathrm{s}}$, the current presentation is admitted, and its frame is weak. Apply the first local input at this prefix. It forms the regular coefficient from the actual companion at level $k+1$, using the old θ_k, x_k and the already available remainder data. By the second input, with $s_j = jc_0 - \sum_{l=1}^{j} b_l^0$, we have $|s_j| \le c_2$. For each candidate segment ending at the computed value $x_{k+1} := x_k - b_{k+1}$, telescoping gives

$$x_i - x_{k+1} - c_0(k+1-i) = s_i - s_{k+1} + \sum_{l=i+1}^{k+1} (f_l + q_l). \tag{3.ix}$$

The input for the last error is x_k, so it obeys eq. 3.iv before any test on x_{k+1}. The earlier errors obey the same bound by induction. This gives the lower inequality in eq. 3.vii. The one-step partial-sum estimate gives $|b_l^0| \le c_0 + 2c_2$. Thus $|b_l^0 + f_l| \le C_\beta$ and $b_l \le B_{\mathrm{s}}$; summing gives the upper bound.

In particular $x_{k+1} \ge x_k - B_{\mathrm{s}} > h$. The new entry remains above the floor and extends the strong and weak frame inequalities. Its tube deviation is zero when $\theta_{k+1} = x_{k+1}$. If the running minimum drops, its drop is at most B_{s}. The cell condition in lemma 3.2 makes the ratio of $(\log y)^r$ at the two successive minima less than L. Taking the least L-powers gives

$$R_j / R_{j+1} \in \{1, L\}. \tag{3.x}$$

The same argument with a bound $2B_{\mathrm{s}}$ on the drop of the minimum across two steps excludes two consecutive drops of R_j. Indeed, two drops by L would force a ratio greater than L between the unrounded powers. Thus the numerical frame, tube and cell requirements have been verified for the proposed next entry. Use the one-step cofinal admission to choose tori and cutoffs on which all reads of this finite extension are admitted. The common-read identity preserves the coefficients already computed when passing to a larger realization. Apply closure on each admitted geometry to obtain the full successor presentation and its new final remainder. The scalar update eq. 2.iv is then evaluated for that realization. It was unnecessary for computing the new weight. Repeating the one-step argument requires no advance choice of the stopping horizon. On any admitted fixed finite torus and cutoff, the same inputs realize the common coefficient prefix through every available step.

If stopping never occurs, this procedure constructs a prefix of every length with $x_j > h + B_{\mathrm{s}}$. The lower drift yields $x_j \le \beta + c_5 - \lambda_{\mathrm{s}} j$, a contradiction for large j. If $K > 0$, its preceding value is above $h + B_{\mathrm{s}}$, and the last decrement is at most B_{s}. This gives the terminal lower endpoint. The upper endpoint follows from the definition of K. Applying the upper drift to the whole run and the lower drift up to $K-1$ gives eq. 3.viii. At $K = 0$, $\beta = h + B_{\mathrm{s}}$ and both bounds hold directly. The lower bound on K then gives the unboundedness assertion. □

Only the block drift in eq. 3.vii is positive at large separations. For example the arithmetic model $b_l^0 = c_0 + (-1)^l d$ with $d > c_0$ has bounded one-loop partial sums after subtracting c_0 and a negative decrement at every odd step. It satisfies the numerical premises with vanishing corrections. Thus the stopping proof cannot require stepwise positivity. For $g = h^{-1/2}$ define the sets

$$\mathcal{W}_k(g) = \{\beta \ge h + B_{\mathrm{s}} : K(\beta, h) = k\}.$$

Unbounded stopping depths make $\mathcal{W}_k(g)$ nonempty for infinitely many k. The existence of one tuned value for every prescribed depth uses a separate argument.

Proposition 3.3 (Finite realization and the torus floor)**.** Let the prefix of theorem A be given by its common companion. If $K' \geq K$, the numerical period conditions at levels $0, \dots, K$ follow from

$$\ell \geq 2L^{11} M R(h + B_{\mathrm{s}}).$$

If the stronger prospective condition stated after eq. 3.v is required at every step, it suffices to multiply this terminal side floor by L. In a finite realization with the supplied actual geometric and analytic admissions, the finite construction then realizes the same weights through K. ■

Proof. By eq. 3.x, $R_i \leq L^{K-i} R_K$. The running minimum is at most x_K, so $R_K \leq R(h + B_{\mathrm{s}})$. Since $P_i = \ell L^{K'-i} \geq \ell L^{K-i}$, the first assertion follows. For the prospective condition at $i+1$, one needs $P_{i+1} \geq 2L^{11} M R_i$, which is obtained with the stated extra factor. These computations select numerical periods. To realize the known run, apply the common-read input and the same closures on the admitted torus and cutoff. The vacuum subtraction still uses the scalar numbers of that realization. □

4. Depth estimates and the logarithmic law

Proposition 4.1 (Terminal-depth bounds)**.** In the trajectory of theorem A, put $x_{[s]} = x_{K-s}$ for $0 \leq s \leq K$. Then

$$h + \max\{0, \lambda_{\mathrm{s}} s - c_5\} < x_{[s]} \leq h + B_{\mathrm{s}}(s+1). \tag{4.i}$$

In addition,

$$h - c_5 + \tfrac{1}{2} c_0 s < x_{[s]} \leq h + B_{\mathrm{s}} + c_5 + \tfrac{3}{2} c_0 s.$$

At fixed h, any sequence with $K \to \infty$ has a subsequence with a limit of $x_{[s]}$ for every fixed s. These limits lie in the closed versions of the intervals, and the corresponding couplings converge by continuity of $x \mapsto x^{-1/2}$. The limiting terminal coupling lies in $[(h + B_{\mathrm{s}})^{-1/2}, h^{-1/2}]$.

For any $H \geq h$, the number of steps $1 \leq j \leq K$ whose preceding value satisfies $x_{j-1} \leq H$ is bounded by $1 + (H - h + c_5)/\lambda_{\mathrm{s}}$, independently of the initial weight. ■

Proof. Apply eq. 3.vii from $K - s$ to K, and use the terminal window. Each preceding value is itself above h. This gives eq. 4.i. For the additional upper bound, the telescoping identity gives $x_{[s]} - x_K \leq c_5 + \frac{5}{4} c_0 s$. The lower bound follows from eq. 3.vii. Each fixed-depth interval has compact closure. Extract successively at depths $0, 1, \dots$, discarding the finitely many runs too short for the depth being considered, and take a diagonal subsequence.

If a preceding value at most H occurs, let i be its first index. Every counted step has preceding index between i and $K - 1$. The lower drift from i to K gives $\lambda_{\mathrm{s}}(K - i) - c_5 \leq x_i - x_K < H - h$. This bounds the entire remaining length, even if the weights later rise above H again. If no such index exists, the count is zero. □

Lemma 4.2 (Sums of running powers)**.** Let $g_j = x_j^{-1/2}$. On the prefix of theorem A, for $p > 2$,

$$\sum_{j=0}^{k} g_j^p \leq C_p \left(1 + \frac{1}{\lambda_{\mathrm{s}}(p-2)}\right) x_k^{1-p/2} \qquad (k \leq K). \tag{4.ii}$$

For the marginal power at $k = K$,

$$\frac{1}{B_{\mathrm{s}}} \log\!\left(1 + \frac{B_{\mathrm{s}}(K+1)}{h + B_{\mathrm{s}}}\right) \leq \sum_{j=0}^{K} g_j^2 \leq \frac{2}{h} + \frac{1}{\lambda_{\mathrm{s}}} \log\!\left(1 + \frac{2\lambda_{\mathrm{s}} K}{h}\right)$$

once $h \geq 2c_5$. For every fixed $p \geq 0$ and $\eta > 0$, let $t_p(y) = y^{-1/2} (\log y)^p$. Above a fixed floor,

$$t_p(x_n) \leq (1+\eta)\sqrt{n-m}\, t_p(x_m), \qquad t_p(x_m) \leq (1+\eta) t_p(x_n) \qquad (m < n).$$ ■

Proof. For $j \le k$, the lower drift gives $x_j \ge x_k - c_5 + \lambda_s(k-j)$. With the fixed floor $x_k \ge \max\{1, 2c_5\}$, this is at least $x_k/2 + \lambda_s(k-j)$. Bound the first term separately and compare the remaining decreasing sequence with the integral on the nonnegative half-line. This yields eq. 4.ii. For $p = 2$ the same integral up to K gives the upper logarithm. The upper terminal-depth bound gives the lower logarithm by integrating the reciprocals of $h + B_s(s+1)$ from zero to $K+1$.

The profile t_p decreases above e^{2p}. If $x_m \ge x_n$, the first ratio is at most $(1 + B_s(n-m)/x_n)^{1/2}$, since its logarithmic factor is at most one. It is at most $(1+\eta)\sqrt{n-m}$ above a fixed floor. If $x_m < x_n$, the first ratio is at most one. In the latter case the reverse ratio is bounded by $(1 + c_5/x_m)^{1/2}$ from the lower drift, again with logarithmic factor at most one. In the former case the reverse ratio is at most one by monotonicity. □

A weighted age sum used for inherited remainder coefficients is also a consequence. For $\kappa_0 > 6$ and $k \le K$,

$$\sum_{j=0}^{k} (x_j/x_k)^2 g_j^{\kappa_0} = x_k^{-2} \sum_{j=0}^{k} g_j^{\kappa_0 - 4} \le C g_k^{\kappa_0 - 2}. \tag{4.iii}$$

Use eq. 4.ii with $p = \kappa_0 - 4 > 2$. This estimate sums an already supplied local remainder bound; it does not establish that bound. The positive block drift, rather than a bound on backward excursions alone, controls the number of contributing levels.

Theorem B (Uniform ultraviolet asymptotics)**.** Set

$$\omega_*(y) = \sup_{t \ge y} \{\epsilon_u(t) + \epsilon_m(t)\},$$

on the half-line of admitted real inverse couplings. For the trajectory in theorem A, the depth error is bounded by

$$|x_{[s]} - x_K - c_0 s| \le \Delta_h(s), \qquad \Delta_h(s) = c_5 + \sum_{u=0}^{s-1} \omega_*(h + \max\{0, \lambda_s u - c_5\}). \tag{4.iv}$$

The envelope is nondecreasing in s and $\Delta_h(s)/s \to 0$ at fixed h. In particular the mean of $|f_j + q_j|$ over the full run tends to zero. At $a_K = L^{-K}$, for $K \ge 1$,

$$\left| \frac{\beta}{\log(a_K^{-1})} - \frac{11}{3\pi^2} \right| \le \frac{h + B_s + \Delta_h(K)}{K \log L}.$$

In particular this convergence is uniform over the admitted initial weights with the same stopping depth and fixed h. If the power and logarithmic envelopes of eq. 3.i hold with $p_R > 1$, then above a fixed floor one may take

$$\Delta_h(s) \le C \left\{ 1 + \sqrt{h + \lambda_s s}\, [\log(h + \lambda_s s)]^{\mathfrak{p}_0} \right\}.$$

The corresponding relative error at fixed h is $O_h(K^{-1/2}(\log K)^{\mathfrak{p}_0})$ as $K \to \infty$. ■□

Proof. Apply the telescoping identity on the last s steps. The one-loop sum error is at most c_5. The preceding argument of each error in those steps has depth u between 1 and s. By eq. 4.i and the monotonicity of ω_*, the error sum is at most the same sum at depths $0, \dots, s-1$. This proves eq. 4.iv. Its summands tend to zero as the depth increases, so their averages tend to zero. At $s = K$, use $x_{[K]} = \beta$ and the terminal window, then divide by $K \log L$. The numerical coefficient is $c_0/\log L$ from the one-loop input with $N = 2$.

Above a fixed monotonicity floor the two errors have a common majorant $Cy^{-1/2}(\log y)^{\mathfrak{p}_0}$. Use $h + \max\{0, \lambda_s u - c_5\} \ge (h + \lambda_s u)/2$ for $h \ge 2c_5$. Bound all logarithms by $\log(h + \lambda_s s)$. The remaining decreasing sum is at most

$$h^{-1/2} + \int_0^s (h + \lambda_s t)^{-1/2}\, dt \le h^{-1/2} + 2\lambda_s^{-1} \sqrt{h + \lambda_s s}.$$

This proves the quantitative estimate with constants independent of the prefix and h above the fixed floor. □

Lemma 4.3 (Change of lattice units)**.** For compatible finite lattices, let b be an integer with $K' - b \geq 0$. In four dimensions,

$$Z_{K',\ell,\beta} = Z_{K'-b,L^b\ell,\beta}.$$

At a fixed initial weight, a datum at horizon $K' \geq K(\beta, h)$ therefore represents the same finite Wilson integral as the first-passage datum at side $L^{K'-K}\ell$. The reverse change of units is valid for the bare integral on the contracted torus. To use the normalized history estimates there one must still meet its geometric admission conditions. ■

Proof. The map $u \mapsto L^b u$ bijects the sites of the two lattices, their positive bonds, and their oriented plaquettes. Give corresponding bonds the same group variables. The plaquette products and normalized product Haar measure are unchanged, and the Wilson action has no spacing prefactor in four dimensions. This proves the equality. When $K' \geq K$, take $b = K' - K$. Identification of intermediate renormalized coefficients uses the common-prefix hypothesis and proposition 3.3, not this bare equality alone. □

Proposition 4.4 (Common prefixes under change of units)**.** Suppose the actual supply rule is natural under the bond relabeling of lemma 4.3. The identified frame and weight prefixes, towers, partitions and retained records must identify the local formulas, Borel tests, measures and determining collars. On the resulting admitted tori and cutoffs, relabeling identifies the formed presentations through their common prefix. For the affine reference frame $\theta_j = h + c_0(K' - j)$, its prefix ending at $K' - s$ has the same frame as the shorter run with terminal reference value $h_s = h + c_0 s$. It is carried to the torus of side $L^s\ell$, for every integer $0 \leq s \leq K'$. A reference side floor of the form $\ell \geq CMR(h + B_{\mathrm{s}})$ implies $L^s\ell \geq CMR(h_s + B_{\mathrm{s}})$ above a fixed floor on h. ■

Proof. The bare measures agree by lemma 4.3. At each common step, transport the actual tests and kernels by the same relabeling. A test indicator and its transported version have the same value on corresponding configurations. A draw and any projection have the corresponding pushforward law by eq. 2.ii. The local densities and companions then agree on identified reads, so their coefficients agree. Induction identifies the retained history measures and the given local expressions at each step in the prefix.

For the affine frame, $h + c_0(K' - j) = h_s + c_0(K' - s - j)$ for $j \leq K' - s$. The relabeling with $b = s$ gives the stated torus. For large y, the ratio $(\log(y + c_0))^r/(\log y)^r$ is less than L. Thus $R(y + c_0) \leq LR(y)$. Iterate this bound s times at $y = h + B_{\mathrm{s}}$ to obtain the size implication. □

For a self-calibrated run, a smaller terminal inverse coupling $h' \leq h$ gives a lower stopping threshold. If $h' \geq Y$ and the prolonged geometries are admitted, the same construction continues the old prefix until its first passage at h'. If the old terminal value already lies at or below $h' + B_{\mathrm{s}}$, the two stopping indices agree. In either case the earlier prefix measures and coefficients are the original ones. Comparing two different histories or frames from a bare change of units alone requires extra compatibility.

5. Tuning with fixed choices

Let $c_0 > 0$ be the one-loop drift in the coupling normalization of this manuscript. Fix a nonnegative integer horizon K and an inverse-coupling floor $h > 1$. For the slope estimates, fix an admitted real frame satisfying, for constants $\lambda_\theta > 0$ and $d_\theta \geq 0$,

$$\theta_j \geq h, \qquad \theta_j - \theta_l \geq \lambda_\theta(l - j) - d_\theta \quad (0 \leq j \leq l \leq K). \tag{5.i}$$

We take $h \geq 2d_\theta$ and increase its floor below. Constants in the slope estimates may depend on the fixed drift and analytic parameters, but not on the horizon or volume. The canonical tuning

interval will use the affine frame with $h = g^{-2}$,

$$\theta_j = h + c_0(K - j), \qquad 0 \le j \le K. \tag{5.ii}$$

Hold the frame and all lattice, cutoff, gauge and scaling choices fixed as the bare weight β varies. Write x_j^θ for its actual weights, $x_0^\theta = \beta$, and $b_i = x_{i-1}^\theta - x_i^\theta$ for its extracted increments. The frame and the actual weights are different inputs to a step.

We use the coefficient comparison of [1, Theorem A and Corollary 6.4] on the actual admitted reads. Its two estimates concern different parameter domains. At step i write

$$b_i = b_i^0 + f_i(z_0, \dots, z_{i-1}) + q_i(\beta), \qquad z_j = \theta_j - x_j^\theta(\beta). \tag{5.iii}$$

The Gaussian numbers b_i^0 are fixed by the reference data and obey $|s_k| \le c_2$, where $s_k = kc_0 - \sum_{i=1}^k b_i^0$. Fix an admitted relative radius $0 < \sigma_* \le 1/2$. The unmarked function f_i is holomorphic on the full product $|z_j| < \sigma_* \theta_j$ for $j < i$, with supremum at most

$$\epsilon_{i-1} = C_{\mathrm{u}} \theta_{i-1}^{-1/2} (\log \theta_{i-1})^{\mathfrak{p}_0}.$$

For the bare-disc route fix $\varrho > 0$, $c_\beta > 0$ and $p_R > 1$. At each admitted real centre, q_i has an actual bare-variable holomorphic germ on the disc of radius $c_\beta \varrho$ about that centre, with supremum at most $\epsilon'_{i-1} = C_{\mathrm{m}} \theta_{i-1}^{-p_R}$. This is the marked coefficient of the same actual step, including the variation of all its supplied inherited remainder data. The marked estimate is a separate input on its disc. All constants in these inputs are uniform in the horizon and admissible torus. The coefficient reads use the common companions and are real on real data.

Lemma 5.1 (A product-disc derivative bound)**.** Let f be holomorphic and bounded by E on the finite product $|z_j| < r_j$, with all $r_j > 0$. Throughout this product,

$$\sum_j r_j (1 - |z_j/r_j|^2) |\partial_j f(z)| \le E. \tag{5.iv}$$

On the half product, $\sum_j r_j |\partial_j f(z)| \le 4E/3$ and $|f(z) - f(\widetilde{z})| \le (4E/3) \max_j |z_j - \widetilde{z}_j|/r_j$. The constant is independent of the number of factors. ■

Proof. Put $\alpha_j = z_j/r_j$. For $|t| < 1$ map each coordinate by

$$\phi_j(t) = r_j \frac{\alpha_j + e^{i\vartheta_j} t}{1 + \bar{\alpha}_j e^{i\vartheta_j} t}.$$

This map takes the disc into the jth factor and has derivative $r_j(1 - |\alpha_j|^2) e^{i\vartheta_j}$ at zero. Choose the phases so that the nonzero derivative terms in $(f \circ \phi)'(0)$ have the same argument. Cauchy's bound on the unit disc gives $\sum_j r_j (1 - |\alpha_j|^2) |\partial_j f(z)| \le E$. On the half product each factor $1 - |\alpha_j|^2$ is at least $3/4$. Integrating the directional estimate on the segment in that convex product proves the difference bound. □

Proposition 5.2 (Slope along a frozen prefix)**.** Use the fixed frame eq. 5.i. Suppose the actual increments on a prefix satisfy eq. 5.iii and the stated analytic bounds. At the real centre in question assume $|z_j| \le \sigma_* \theta_j/2$ for all used entries. The earlier weights and the last formed coefficient are germs of the actual functions at this same centre. Then, uniformly in prefix length and volume,

$$\sum_{i=1}^{k} |b_i'| \le e^{A_k}(1 + D_k) - 1, \qquad A_k \le C h^{-1/2} (\log h)^{\mathfrak{p}_0}, \quad D_k \le C \varrho^{-1} h^{1-p_R}, \tag{5.v}$$

where primes denote differentiation in β. For sufficiently large h the right side is at most $1/3$. Thus $2/3 \le (x_j^\theta)' \le 4/3$ on real admitted prefixes. On a real segment with these hypotheses throughout, the individual increments have Lipschitz bounds $(4/3)a_i + d_i$, with a_i, d_i as in eq. 5.vi. Their sum is bounded by $C\{h^{-1/2}(\log h)^{\mathfrak{p}_0} + \varrho^{-1} h^{1-p_R}\}$. ∎

Proof. In the unmarked term, $z_j' = -(x_j^\theta)'$. For $j < i$, the frame bound gives $\theta_j \ge \theta_{i-1} - d_\theta \ge \theta_{i-1}/2$. Lemma 5.1 and Cauchy's bound on the actual marked disc give

$$|b_i'| \le a_i M_{i-1} + d_i, \qquad M_{i-1} = \max_{0 \le j < i} |(x_j^\theta)'|, \quad a_i = \frac{8\epsilon_{i-1}}{3\sigma_* \theta_{i-1}}, \quad d_i = \frac{2\epsilon_{i-1}'}{c_\beta \varrho}. \tag{5.vi}$$

The factor two in d_i allows Cauchy circles strictly inside the supplied disc. This differentiation includes the actual inherited remainders through q_i.

Since $(x_0^\theta)' = 1$, the differentiated scalar recursion bounds M_{i-1} by $V_{i-1} = 1 + \sum_{l<i} |b_l'|$. Then $V_i \le (1 + a_i) V_{i-1} + d_i$. Iterating gives $V_k \le e^{A_k}(1 + D_k)$ for $A_k = \sum_{i \le k} a_i$ and $D_k = \sum_{i \le k} d_i$. For example, each product of factors $1 + a_i$ is bounded by the exponential of their sum. For $i \le k$, eq. 5.i gives

$$\theta_{i-1} \ge h + \lambda_\theta (k - i) - d_\theta \ge \tfrac{1}{2}\{h + \lambda_\theta (k - i)\}.$$

Above a fixed floor the summands decrease. Comparison with the integrals along this affine lower envelope bounds the first sum by $C h^{-1/2} (\log h)^{\mathfrak{p}_0}$ and the second by $C \varrho^{-1} h^{1-p_R}$. For the first integral, integrate by parts and absorb the term with exponent $\mathfrak{p}_0 - 1$ at large h. The second is the integral of a power with exponent less than -1. Both bounds tend to zero. Finally $(x_j^\theta)' - 1 = -\sum_{i \le j} b_i'$ gives the claimed slopes. Substitute $M_{i-1} \le 4/3$ in eq. 5.vi and integrate over the admitted segment to obtain the Lipschitz bounds. Their sum is at most $(4/3)A_k + D_k$. □

The relative disc for f_i supplies one inverse power of θ_{i-1}. The bare disc for q_i supplies one inverse power of ϱ. The total slope error in eq. 5.v is $O(h^{-1/2}(\log h)^{\mathfrak{p}_0} + \varrho^{-1} h^{1-p_R})$. In particular the first term controls its order when $p_R \ge 3/2$ and ϱ is fixed. The derivative bound for a product disc gives no decay with the age of a particular input coordinate.

Proposition 5.3 (Slope with an independent marked polydisc)**.** Use the fixed frame eq. 5.i and the stated unmarked coefficient bound. Suppose instead that the actual marked coefficient factors through the earlier deviations in eq. 5.iii as a holomorphic function $q_i(z_0, \dots, z_{i-1})$. Assume that it has supremum at most $C\theta_i^{-p_R}$ on the product of discs of radii $c_{\mathrm{m}} \theta_j (\log \theta_j)^{-C_*}$, $j < i$, for fixed $c_{\mathrm{m}} > 0$, $C_* \ge 0$ and $p_R > 0$. This factorization includes the genuine inherited dependence. Assume the runs remain in the half products for both terms, with the same compatible core and coefficient choices. Then the bare slope satisfies

$$\sum_{i \le k} |b_i'| \le e^{A_k^\flat} - 1, \qquad A_k^\flat \le C\big[h^{-1/2}(\log h)^{\mathfrak{p}_0} + h^{-p_R}(\log h)^{C_*}\big]. \tag{5.vii}$$

The bound is uniform in K, k and compatible volume. ∎

Proof. Above a fixed floor $y(\log y)^{-C_*}$ increases. For $j < i$, the frame bound gives $\theta_j \ge \theta_i - d_\theta \ge \theta_i/2$. The marked radii are therefore bounded below by a fixed multiple of $\theta_i (\log \theta_i)^{-C_*}$. The unmarked

radii are bounded below by $\sigma_*\theta_{i-1}/2$, and $\theta_{i-1} \geq \theta_i/2$. Using eventual monotonicity of $y^{-3/2}(\log y)^{\mathfrak{p}_0}$ in the unmarked bound, lemma 5.1 gives the absolute derivative row bound

$$a_i^\flat \leq C\big[\theta_i^{-3/2}(\log\theta_i)^{\mathfrak{p}_0} + \theta_i^{-p_R-1}(\log\theta_i)^{C_*}\big].$$

The chain rule gives $|b_i'| \leq a_i^\flat M_{i-1}$. The scalar iteration in the preceding proof with $d_i = 0$ gives $1+\sum_{i\leq k}|b_i'| \leq \exp(\sum_{i\leq k} a_i^\flat)$. Here $\theta_i \geq \{h+\lambda_\theta(k-i)\}/2$. Compare both sums with the integrals of their decreasing envelopes above $h/2$. Writing $y = (h/2)u$ leaves an integrable power $u^{-1-\delta}$ times a fixed power of $1+\log u$, with $\delta > 0$. This gives eq. 5.vii. □

5.1. **The interval of admitted bare weights.** Let $\mathsf{t}^\sharp(y) = \varpi y(\log y)^{-2q_\mathrm{t}}$ with $\varpi > 0$ and $q_\mathrm{t} \geq 0$. A tube is a positive nonincreasing sequence t_j, with $\mathsf{t}_j \leq \mathsf{t}^\sharp(\theta_{j+1})$ for $j < K$. For $K = 0$ use $\mathsf{t}_0 \leq 2\mathsf{t}^\sharp(\theta_0)$. For any frame of length $K \geq 1$ the largest tube has entry $\min_{1\leq i\leq \min(j+1,K)} \mathsf{t}^\sharp(\theta_i)$ at j. The preceding upper bounds and monotonicity require this minimum, and these minima themselves satisfy the requirements. A constant tube of radius t is permitted whenever $t \leq \varpi x_*^{3/4}/C_a$ and every frame entry is at least $x_* > e$, where $a = 2q_\mathrm{t}$ and $C_a = \max\{1, (4a/e)^a\}$ with $C_0 = 1$ in this formula. Indeed, maximizing $u^a e^{-u/4}$ gives $(\log y)^a \leq C_a y^{1/4}$ for $y \geq 1$.

For the canonical interval construction, now specialize to the affine frame eq. 5.ii. Put $\iota_j = \epsilon_j + \epsilon_j'$ and choose

$$T_K = \varrho + c_2, \qquad T_j = T_K + \sum_{l=j}^{K-1} \iota_l, \qquad \widehat{\mathsf{t}}_j = T_j + c_2 + 8\varrho, \tag{5.viii}$$

$$F_j(\beta) = \theta_j - x_j^\theta(\beta) + s_j.$$

Here F_j is defined once the weight at j is computed, before requiring closure at that weight. The sum of ϵ_l in T_j is at most $C\theta_j^{1/2}(\log\theta_j)^{\mathfrak{p}_0}$. To see this, bound its logarithm factor by the one at θ_j and sum the inverse square roots on the reference line by an integral. The sum of ϵ_l' is at most Ch^{1-p_R}. For $j < K$ and sufficiently large h, $\theta_{j+1} \geq \theta_j/2$. Consequently the ratio of $\widehat{\mathsf{t}}_j$ to $\mathsf{t}^\sharp(\theta_{j+1})$ tends uniformly to zero as h increases. The terminal value of $\widehat{\mathsf{t}}$ is fixed, and the one-entry convention holds at the same large floor. The sequence in eq. 5.viii is therefore a tube. Increasing the floor also makes $\widehat{\mathsf{t}}_j \leq \sigma_*\theta_j/2$ at every level. At retained entries $x_j^\theta \geq \theta_j - T_j - c_2 \geq \theta_j/2$. Choose $h/2$ above all absolute input floors. The terminal range $[h-\varrho, h+\varrho]$ then also stays above those floors.

Theorem C (Full tuning in the canonical interval)**.** Use the reference frame eq. 5.ii, the actual increment bounds of eq. 5.iii, and the uniform one-loop partial-sum bound above. Assume the following admitted prefix operation. The initial Wilson inputs are admitted on the interval $|\theta_0-\beta| \le T_0$. From each real prefix with $|F_j| \le T_j$ at every closed level, the actual next coefficient is formed on a neighbourhood before imposing the new bound on F_{j+1}. The coefficient and all earlier weights there have the analytic inputs of proposition 5.2. When the computed entry satisfies $|F_{j+1}| \le T_{j+1}$, the actual next level closes with the declared tube room. This operation includes the inherited history and vacuum data at each closed prefix. It uses one frame, with common companions and coefficients on the admissible tori.

For a floor on h independent of K, there are nested compact intervals I_j of actual closed prefixes with $|F_l| \le T_l$ for $l \le j$. On I_j, F_j decreases from T_j to $-T_j$. For each real v with $|v-h| \le \varrho$ there is one $\beta \in I_K$ such that $x_K^\theta(\beta) = v$. The tuned bare weight β_K^θ for $v = h$ satisfies

$$\tfrac{1}{2}c_0K - 2c_2 \le \beta_K^\theta - h \le \tfrac{3}{2}c_0K + 2c_2, \qquad |\beta_K^\theta - h - c_0K| \le c_2 + \sum_{j<K} \iota_j. \tag{5.ix}$$

For all $\beta \in I_K$ and $0 \le i \le j \le K$ one has

$$\tfrac{1}{2}c_0(j-i) - 2c_2 \le x_i^\theta - x_j^\theta \le \tfrac{3}{2}c_0(j-i) + 2c_2.$$

The uniqueness assertion concerns the fixed frame and its canonical interval. ■□

Proof. The initial $F_0 = \theta_0 - \beta$ maps I_0 decreasingly onto $[-T_0, T_0]$. The high floor makes all initial weights positive. At a closed prefix the bound $|F_l| \le T_l$ gives $|z_l| \le T_l + c_2$, leaving room 8ϱ inside the chosen tube. The premise supplies the next coefficient before any closure at the next weight. Its real function F_j on a neighbourhood of I_{j-1} has derivative $-(x_j^\theta)' \le -2/3$ by proposition 5.2. Moreover

$$F_j - F_{j-1} = b_j - b_j^0, \qquad |F_j - F_{j-1}| \le \iota_{j-1}. \tag{5.x}$$

This follows directly from the changes in the frame and in s_j. At the left endpoint of I_{j-1}, $F_{j-1} = T_j + \iota_{j-1}$, so $F_j \ge T_j$. At the right endpoint $F_j \le -T_j$. The strict slope and continuity give the unique ordered preimages of T_j and $-T_j$. Their interval I_j meets the new bound at every point. The supplied closure operation applies on it with the same room.

At K, $x_K^\theta = h + s_K - F_K$. Its image contains $[h-\varrho, h+\varrho]$ because $T_K = \varrho + c_2$ and $|s_K| \le c_2$. Strict monotonicity gives the stated uniqueness. Summing the increments from $i+1$ through j gives

$$x_i^\theta - x_j^\theta = c_0(j-i) + s_i - s_j + \sum_{i<l\le j} (b_l - b_l^0).$$

The high floor gives $\sup_l \iota_l \le c_0/2$. This proves the spacing bounds. At the tuned value $x_K^\theta = h$ and $s_0 = 0$, the same sum gives eq. 5.ix. For $K = 0$, the initial interval and $x_0^\theta = \beta$ already prove all assertions. □

On I_K the terminal slope gives, for a bare offset t with $\beta_K^\theta - t \in I_K$,

$$x_K^\theta(\beta_K^\theta - t) = h - t + O(\eta_h|t|), \qquad \eta_h = e^{A_K}(1+D_K) - 1.$$

This follows by integrating the slope error on the intervening real segment. The tuned endpoint h is a condition on the fixed reference run. First passage is defined separately for the self-calibrated trajectory.

6. A backward equation for homogeneous profiles

Use the complete profile spaces and the scalar interface of [1, Theorem B and Corollary 7.5]. For a prescribed weight sequence x, the forward profile $\psi_j(x)$ starts at zero. At each step the actual homogeneous formation carries the scalar $x_j - b_{j+1}^{\mathrm{G}} - r_j(x_j, \psi_j(x))$. Here b_{j+1}^{G} is the companion coefficient of the actual reference Gaussian, and r_j is the bounded companion readout of the actual

unprojected formation. The Gaussian coefficient is fixed when the action weights vary. For an endpoint reference X at level n, the nominal frame at level j is $a_j = X + \sum_{\ell=j}^{n-1} b^{\mathrm{G}}_{\ell+1}$. On the declared relative weight box about this frame the forward result supplies

$$\begin{aligned} |r_j(x_j, \psi_j(x))| &\le Ce(a_j), \\ |r_j(x_j, \psi_j(x)) - r_j(\widetilde{x}_j, \psi_j(\widetilde{x}))| &\le Ck(a_j)\|x - \widetilde{x}\|_\infty, \\ \|\psi_j(x) - \psi_j(\widetilde{x})\|_{j,*} &\le Ck(a_j)\|x - \widetilde{x}\|_\infty. \end{aligned} \tag{6.i}$$

The functions are $e(y) = y^{-1}(\log y)^m$ and $k(y) = e(y)/y$ with fixed $m > 1$. The first two bounds hold for $0 \le j < n$ and concern the unprojected scalar interface. The profile difference bound holds for $0 \le j \le n$, including the terminal profile. These are inputs to this construction.

Theorem D (Homogeneous terminal tuning)**.** Fix a horizon $n \ge 0$, a positive real endpoint reference X, and real Gaussian coefficients with

$$\left| \sum_{\ell=j}^{n-1} (b^{\mathrm{G}}_{\ell+1} - b_\infty) \right| \le B_a, \qquad 0 \le j \le n,$$

for fixed $b_\infty > 0$ and $B_a < \infty$. Define

$$a_j = X + \sum_{\ell=j}^{n-1} b^{\mathrm{G}}_{\ell+1}, \qquad a_n = X.$$

Assume the admitted actual forward profile and scalar operations of eq. 6.i on the frozen frame a, with holomorphy and the uniform profile bound on the buffered domains of the forward theorem. Their exact physical exponent includes the scalar vacuum factor as a separate number. Fix a relative radius $\delta \in (0, 1/2]$ within the inner profile weight domain.

Above a floor on X independent of the horizon, for every complex ζ with $|\zeta| < cX$ for a fixed $c > 0$, there is one trajectory in the box $|x_j - a_j| \le \delta a_j$ with initial profile zero and terminal weight $x_n = X + \zeta$. It is real for real data and real ζ. The trajectory and its forward profiles are holomorphic in ζ. At $\zeta = 0$, $x_j \asymp X + n - j$ and

$$x_0 = X + b_\infty n + O\big(1 + (\log(X + n))^{m+1}\big). \tag{6.ii}$$

All these constants are uniform in n and the admitted volume. On a fixed smaller endpoint disc,

$$\sup_j |x_j' - 1| \le C\frac{(\log X)^m}{X}, \qquad \|\psi_j'\|_{j,*} \le Ck(a_j), \qquad |x_j^{(s)}| \le C_s\frac{(\log X)^m}{X^s} \quad (s \ge 2). \tag{6.iii}$$

In particular the real bare weight is strictly increasing with the real endpoint there. ■□

Proof. The Gaussian bound gives $|a_j - X - b_\infty(n - j)| \le B_a$. In particular $a_j \asymp X + n - j$ above a fixed floor, even if some Gaussian increments are negative. Compare the sums with those over $X + c_1 q$ for $q \ge 0$, using fixed $c_1 > 0$. The first comparison integrates $(\log y)^m/y$ up to Ca_j. The second integrates $(\log y)^m/y^2$ on a half-line, where integration by parts bounds it by $C(\log X)^m/X$. Including the first terms gives

$$\sum_{\ell=j}^{n-1} e(a_\ell) \le C\big(1 + (\log a_j)^{m+1}\big), \qquad \sum_{\ell<n} k(a_\ell) \le C(\log X)^m/X. \tag{6.iv}$$

The backward sum of the exact outgoing weights is the fixed-point equation

$$(\Phi^\zeta x)_j = a_j + \zeta + \sum_{\ell=j}^{n-1} r_\ell(x_\ell, \psi_\ell(x)). \tag{6.v}$$

Take $c \le \delta/8$. Since $a_j \ge X/2$ for large X, the endpoint term is at most $\delta a_j/4$. By the first bound in eq. 6.iv, increasing the floor makes the remainder sum at most $\delta a_j/4$. Thus the image is

within the half box. The second bound in eq. 6.iv and the scalar difference bound in eq. 6.i give a contraction in the unweighted supremum norm with rate $\mathfrak{k}_X \le C(\log X)^m/X$. Increase the floor so that $\mathfrak{k}_X \le 1/5$. The closed finite box is complete. Iteration gives one fixed point, and taking adjacent differences in eq. 6.v gives the actual outgoing-weight recursion. Every trajectory in the box has the prescribed forward profiles and must satisfy that equation, which proves uniqueness there.

Iteration starting at a preserves reality on the real interval. For complex ζ the iterates are holomorphic because the prescribed profiles and scalar readout are holomorphic on the larger admitted domain. The uniform contraction gives normal convergence on smaller endpoint discs, so the fixed point is holomorphic there. Summing the scalar correction and using eq. 6.iv gives eq. 6.ii. The exact unprojected step identifies the resulting physical profile and carried weight at each level with its actual integral. The separate scalar vacuum factor is the one of that step.

At the fixed point, which is in the interior of the box, the operator norm of $D_x\Phi^\zeta$ is at most $\mathfrak{k}_X$. Differentiating the fixed-point equation gives $\|x'\|_\infty \le 1 + \mathfrak{k}_X\|x'\|_\infty$ and hence $\|x' - 1\|_\infty \le \mathfrak{k}_X/(1-\mathfrak{k}_X)$. The forward profile estimate gives the claimed bound on ψ_j'. Apply Cauchy's estimate to $x_j' - 1$ on circles of radius a fixed multiple of X inside the larger endpoint disc. Each additional derivative costs a factor X^{-1}, giving the higher jets in eq. 6.iii. The real slopes lie in $[3/4, 5/4]$. When $n = 0$, the sums are empty and the equation is $x_0 = X + \zeta$, with zero profile. □

The actual operations in this theorem are the homogeneous regular ones on their fixed physical radii. Their scalar interface is separate from the actual marked bare-disc dependence used in theorem C.

7. Comparison of admitted presentations

For a real run over a fixed frame, let x_j be the weight used in the Gaussian presentation. A uniform scalar shift c_j is carried in the local action as $\mathsf{c}_j A$. The effective Wilson weight in the all-small-field exponent is $x_j - \mathsf{c}_j$. We consider presentations with one Euclidean regular member at each level, one companion coefficient and one scalar shift, with the real weights and shifts admitted in their stated domains. All tests, charts and underlying integration choices are those of the common frame. In particular the shift is not a spatially varying source.

Write $b_j = x_{j-1} - x_j$ for the chosen decrement, and $\widehat{b}_j$ for the scalar actually read from the new regular member. The bookkeeping identities are

$$\begin{aligned} \mathsf{c}_j &= \mathsf{c}_0 + \sum_{l=1}^{j} (\widehat{b}_l - b_l), \\ x_j - \mathsf{c}_j &= x_{j-1} - \mathsf{c}_{j-1} - \widehat{b}_j. \end{aligned} \tag{7.i}$$

The first line follows by successively equating the effective outgoing coefficient to the incoming coefficient minus the coefficient read from the physical step.

Proposition 7.1 (Matching on common all-small-field windows)**.** Consider two such real presentations, with primed symbols for the second, on one fixed frame. Assume their unnormalized all-small-field measures agree on every admitted torus through level n. At each level m assume an open physical window $\mathfrak{O}_m$ about the identity where the relevant tests equal one and the physical density has its positive continuous all-small-field formula. These windows are hereditary. The earlier averages of a background in $\mathfrak{O}_m$ belong to the corresponding windows. The complete retained physical pairs, including their physical currents, agree at the corresponding earlier reads up to the admitted site gauges. For each outer datum, use the common parametrization of the actual free fluctuations on a connected complex domain containing the real cut and zero fluctuation. The incoming physical exponent and each omitted remainder total, composed with their actual physical reads, are holomorphic there. A nonempty open set in the full real fluctuation coordinate space

maps all those reads into the hereditary windows. These are hypotheses on the actual fluctuation maps, not on a detached independent-current extension.

Assume the exact common formation and vacuum conventions. The centered raw regular integral uses the incoming effective exponent, with the omitted earlier remainder terms removed by their actual common reads. The two Gaussian presentations of its logarithm separate the same reference constants and the respective term $-(D_j/2)\log(x_{j-1}/\theta_{j-1})$ at step j, where D_j is the real dimension of the free fluctuation coordinates. Write $\mathscr{E}_l(P) = \sum_{X,z} \mathcal{E}_l(X,P,z)$ and $\mathscr{R}_l(P) = \sum_X \mathcal{R}_l(X,P)$ for the complete raw totals on the physical pair P. Let $P_l(W)$ be the retained physical pair at the prescribed earlier averaged datum, on which the level-l family is read in the window $\mathfrak{O}_m$. On $\mathfrak{O}_m$ the exact raw all-small-field density is taken in a common reference measure with full support on each open window, such as product Haar measure in the free link coordinates. It has the form

$$p_m(W)\exp\left\{E_{(m)} - (x_m - \mathsf{c}_m)A(U_m(W)) + \sum_{l\le m}\left(\overline{\mathscr{E}}_l(P_l(W)) + \overline{\mathscr{R}}_l(P_l(W))\right)\right\}. \tag{7.ii}$$

Here p_m is the same positive prefactor in both presentations, $\overline{\mathscr{E}}_l$ subtracts the trivial value and its own extracted classical term, and $\overline{\mathscr{R}}_l = \mathscr{R}_l - \mathscr{R}_l(P_\circ)$. The trivial datum is $P_\circ = (\mathbf{1}, 0)$. Require $P_l(\mathbf{1}) = P_\circ$ up to the admitted gauge for every retained read. In particular the actual physical current vanishes at the trivial field, and the trivial background satisfies $A(U_m(\mathbf{1})) = 0$. The complete remainder at m is used after closure at m. The once-counted increment of $E_{(m)}$ in eq. 2.iv contains the displayed Gaussian scalar and $\mathscr{E}_m(P_\circ) + \mathscr{R}_m(P_\circ)$. Its other factors are the same in both presentations. Use cofinal compatible tori and the common physical coefficient readout of [1, Proposition 6.3].

Then $x_j - \mathsf{c}_j = x'_j - \mathsf{c}'_j$ through n. The extracted coefficients match, and on every corresponding hereditary background the regular totals obey

$$\mathscr{E}_j(P) - \mathscr{E}'_j(P) = \frac{D_j}{2}\log(x_{j-1}/x'_{j-1}). \tag{7.iii}$$

The remainder totals and the accumulated vacuum agree through n on their common all-small-field windows. If the next regular formation satisfies the same common-read, cut and continuation hypotheses on a common physical window, its coefficient agrees as well. ■

Proof. At the bare level equality of the positive densities identifies the effective Wilson weight. The action is zero at the identity and is nonconstant on its neighbourhood. On a later open window equality of the measures gives pointwise equality of the continuous densities. Indeed the stipulated common reference measure has full support there. Taking the real logarithm introduces no branch ambiguity.

Proceed by induction. The preceding regular and remainder identities hold on the earlier backgrounds of the new window by heredity. If the formation omits recent remainders from its regular list, their totals already agree at the actual common reads. The composed differences vanish on the stated real open fluctuation patch. Successive one-variable identity theorems make them zero on a complex neighborhood, and connectedness makes them zero throughout the common fluctuation domain. Thus the incoming physical exponents in the next centered regular integral agree. The possibly Borel cut indicators are applied only on the real slice. Its subtraction at zero fluctuation is also identical. Integration on the same real cut gives the same raw regular logarithm. The prescribed scalar difference gives eq. 7.iii. The curvature coefficient of a field-independent total is zero. The cofinal companion readout therefore gives $\widehat{b}_j = \widehat{b}'_j$, and eq. 7.i identifies the new effective weights.

The regular difference in eq. 7.iii cancels after its value at the trivial datum is subtracted. The classical terms agree by equality of the extracted coefficients. At $W = \mathbf{1}$ the right side of eq. 7.ii is $p_m(\mathbf{1})e^{E_{(m)}}$. Equality there identifies the accumulated vacuum. Dividing by the common effective

Wilson, regular and earlier remainder factors identifies the vacuum-subtracted new remainder total. In the prescribed vacuum increment, the change of the Gaussian scalar cancels the change of the regular trivial value by eq. 7.iii, and the other factors are common. Equality of the vacuum increments therefore gives equality of the new remainder trivial values. This finishes the induction. At a next regular formation the comparison through the raw regular integral and coefficient uses only the preceding equal data. It does not require closure at the new weight. □

The use of this result for a constant scalar shift is precise. Take an admitted bare centre β and a real shift ζ on a common admitted interval. Compare the presentation starting at $\beta - \zeta$ with zero initial shift and the presentation using the Gaussian weights $x_j^\theta(\beta)$ of the admitted run at β, with initial shift ζ carried subsequently as ζ_j. If their actual all-small-field measures and reads meet the hypotheses above, then

$$x_j^\theta(\beta - \zeta) = x_j^\theta(\beta) - \zeta_j. \tag{7.iv}$$

The two initial densities coincide since $e^{-\beta A + \zeta A} = e^{-(\beta-\zeta)A}$. When both presentations use the same fixed labelled kernels, each all-small-field measure is obtained by applying the same linear kernel to this density. This supplies the measure equality for the application. It uses the actual identity of the tests, gauge weights and retained maps in each labelled kernel. On an admitted real segment with the stated slope hypotheses, proposition 5.2 gives $|\zeta_j| \le 4|\zeta|/3$. The fixed frame may be the affine reference frame or the weight prefix of the self-calibrated run at the bare centre. In the latter case hold that whole reference prefix fixed while varying the bare weight on the segment. Differentiating eq. 7.iv at $\zeta = 0$ gives $\partial_\zeta \zeta_j|_{\zeta=0} = (x_j^\theta)'(\beta)$. This is the normalization of the running constant insertion in this comparison. Identification with any separate observable construction requires that construction's own actual matching.

The bare-variable holomorphic germs about centres of a canonical interval I_k can be glued when they agree with the actual real run on the real part of that interval. For two distinct centres in I_k, intersection of their open discs implies that their common real intervals contain an open segment between the centres. For coincident centres, use the one-sided interval inside I_k if necessary. The identity theorem on the convex disc intersection gives equality there. For regular and remainder totals, first fix a physical datum in a window common to both germs throughout an open admitted real subinterval of the overlap. Their actual totals agree there, with the fixed scalar convention, so the same argument identifies their bare-variable continuations at that datum on the connected common domain containing that real subinterval. This identification uses the stated common physical window. A comparison at centres from different real components needs a common admitted real interval or a separately supplied identification of the analytic branches. This gluing uses one frame and compares complete physical totals on the stated windows.

Lemma 7.2 (A weighted two-run estimate)**.** Let r_0, n be integers with $0 \le r_0 < n$, let $0 < \rho < 1$ and $\Pi \ge 0$, and set

$$S_w = \frac{\rho^{n-w+1}}{1-\rho} + (w - r_0)\Pi, \qquad r_0 < w \le n.$$

Suppose the nonnegative quantities $y_w, D_u, k_{u,w}, h_u, h'$ of a finite comparison satisfy, for $r_0 \le s, u < n$,

$$y_{s+1} \le \sum_{u=r_0}^{s} D_u, \qquad D_u \le M_*(\rho^{n-u} + \Pi) + \sum_{w=u+1}^{n} k_{u,w} y_w,$$

$$\sum_{w=u+1}^{n} k_{u,w} S_w \le h_u S_{u+1} + h'\Pi, \qquad \sum_{u=r_0}^{n-1} h_u + h' \le 1/2,$$

where $M_* \ge 0$. Then $y_w \le 2M_* S_w$ for every indicated w. ■

Proof. The weights S_w are positive and nondecreasing. Put $Z = \max_{r_0 < w \le n} y_w / S_w$. The forcing sum through s is at most $M_* S_{s+1}$ by the geometric sum. The other sum is bounded by

$$Z\left(\sum_{u=r_0}^{s} h_u S_{u+1} + (s - r_0 + 1)h'\Pi\right) \le \tfrac{1}{2} Z S_{s+1}.$$

Divide by S_{s+1} and take the maximum. This gives $Z \le 2M_*$. □

The kernel inequality in this last lemma is a premise on the actual comparison of two runs. It allows dependence on later depths. A bound on the single-run increments alone supplies no such kernel estimate.

8. The zero-source partition function

Fix an admitted real run through level n. Its terminal measure is μ_n, and the common subtraction $E_{(n)}$ is given by eq. 2.iv. Let $\mathcal{S}$ be a standing label at level n. Histories with that label have already been summed in its measure $\tau_{\mathcal{S}}$. In particular the empty label includes histories whose large-field components have been healed. Positivity and exactness give

$$\mu_n = \sum_{\mathcal{S}} \tau_{\mathcal{S}}, \qquad \tau_{\mathcal{S}} \ge 0, \qquad \sum_{\mathcal{S}} \tau_{\mathcal{S}}(1) = Z_{K',\ell,\beta}. \tag{8.i}$$

Each measure can be projected to the final links for the purposes of this section. The history sum does not require an exponential formula for the measure of a single raw history.

8.1. **Bounding the retained integrals.** The exponent of a label has a numerical part $v_{n,\mathcal{S}}$. Use the sign in which the numerical factor in the measure is $\exp v_{n,\mathcal{S}}$. For the empty label this factor is $\exp E_{(n)}$. For another label the difference $v_{n,\mathcal{S}} - E_{(n)}$ stays in its normalized contribution. The absolute value of this difference need not be bounded by a constant per final site if an old fine region is present.

Lemma 8.1 (Majorizing coupled integrals)**.** Let $\xi_1, \dots, \xi_b$ be disjoint groups of retained variables, with sigma-finite reference measures and product reference measure $\prod_i \mu_i(d\xi_i)$. Let y denote the remaining variables, with a probability reference measure $\pi(dy)$. Suppose the factor $f_i(\xi_i, e_i)$ is evaluated at an admissible exterior argument $e_i = e_i(y, \xi_1, \dots, \xi_b)$. Assume the admitted support is Borel and the evaluated factors are measurable there. Include the support restrictions in the domain of integration. If $\mathcal{D}_i(\xi_i)$ contains all those exterior arguments, then an integral of the absolute product of the factors is bounded by

$$\int \prod_i |f_i(\xi_i, e_i)|\, d\pi(y) \prod_i d\mu_i(\xi_i) \;\le\; \prod_i \int^{*} \sup_{e \in \mathcal{D}_i(\xi_i)} |f_i(\xi_i, e)|\, d\mu_i(\xi_i). \tag{8.ii}$$

Assume each upper integral on the right is finite. The integral on the left is over the admitted support. Here $\int^*$ is the upper integral over nonnegative measurable majorants. The same conclusion holds with a nonnegative Borel factor depending only on ξ_i inserted on both sides for each i, provided the modified upper integrals are finite. ■

Proof. For each i choose a measurable majorant of its supremum, taking the supremum over an empty exterior set to be zero. Drop the support indicator after applying the majorants. The product then depends only on the disjoint groups ξ_i, so Tonelli's theorem gives the product of their integrals. Integrating y costs one. Taking majorants with integrals arbitrarily close to the finite upper integrals proves the estimate used here. The inserted factors can be included in the definition of f_i. □

The same estimate controls composition of retained kernels. For a measurable kernel $K(\xi; p)$, nonnegative Borel site weight $v(\xi)$ and $c \geq 0$, set

$$\|K\|_c = \int^* e^{cv(\xi)} \sup_p |K(\xi; p)| \, d\mu(\xi).$$

If this is finite, the integral operator $(T_K F)(p) = \int K(\xi; p) F(\xi, p) \, d\mu(\xi)$ satisfies $\|T_K F\|_\infty \leq \|K\|_0 \|F\|_\infty$ for bounded Borel F. For two kernels, suppose the actual measurable read $q(\xi, p)$ belongs to the second exterior domain at η for every (ξ, η, p) on the admitted joint support. Extend the evaluated integrand by zero outside that support. Bound the joint supremum pointwise by the two local suprema, then use the product-majorant argument of eq. 8.ii to obtain

$$\int^* e^{c(v_1(\xi)+v_2(\eta))} \sup_p |K_1(\xi; p) K_2(\eta; q(\xi, p))| \, d\mu_1(\xi) d\mu_2(\eta) \leq \|K_1\|_c \|K_2\|_c.$$

Both norms are assumed finite. Discrete record sums with their normalized frame weights can be included in the reference measures. The actual read inclusion is required before making this estimate.

The distinction between the two orders of supremum and integration is necessary. For $0 < \epsilon < 1$, on $[0, 1]^2$ take both component factors to be $1_{\{|u-v|<\epsilon\}}$. At each fixed exterior value the integral in the component's own variable is at most 2ϵ. Their product has integral $2\epsilon - \epsilon^2$, which exceeds $4\epsilon^2$ for $\epsilon < 2/5$. The upper integrals in eq. 8.ii do bound this product.

For a retained operation born on X at level i, write ω for its discrete record, ξ for its stored variables and μ_ω for their reference measure. On its support let $w_\omega(\xi, e)$ be the actual integrand with the energy factors assigned to this operation. Let $\varpi(\omega)$ be the product of the normalized reference-frame weights for this record. Other discrete choices have counting weight one. The supremum in its weighted mass is taken in a determining domain chosen at its birth using only the prefix through level i. The supplied invariant inclusion places every later admitted evaluation in that birth domain. It is not enlarged after a future evaluation has been selected. In a drop-out it is taken over the admitted real evaluations compatible with the stored variables and the recorded tests, except for the test that caused the drop-out. The mass used in the comparison is

$$\mathsf{M}_c(i, X; \mathcal{B}) = \sum_{\omega \in \mathcal{B}} \varpi(\omega) \int^* e^{c\, v(\omega)} \sup_e |w_\omega(\xi, e)| \, d\mu_\omega(\xi). \tag{8.iii}$$

Here $\mathcal{B}$ can restrict the visible records of the operation, and $v(\omega)$ is the sum of $1 + \log x_t$ over its charged stored sites at their own levels t. In this sum each site at a given level is charged at most a fixed number of times. A mass bound is required for the actual reference measures, including their Jacobians or densities when these occur in the step.

An island includes its assigned core Wilson factor. A drop-out uses a share of the Wilson action of the term that carries it, over bare plaquettes in its assigned core, because the retained operation by itself has no such factor. A failure of a plaquette test uses an assigned positive Wilson contribution; a failure of an approximate fluctuation test may also use a retained gauge contribution. The

genuine fluctuation, collar and rim tests use the corresponding positive quadratic contributions on their carriers. More precisely, if a certified failure forces an allocated nonnegative energy $Q \geq 2a$ on the test support, then

$$1_{\{\text{failure}\}} e^{-Q} \leq e^{-a} e^{-Q/2}.$$

The reference integral of the remaining density must still be estimated. For example replacing e^{-Q} by $e^{-Q/2}$ in a full-space d-dimensional positive quadratic Gaussian integral multiplies its mass by $2^{d/2}$. This follows by scaling all d coordinates by $\sqrt{2}$ and is a volume cost to be recorded. For a site-gauge factor normalized by the truncated Haar number $z(y)$ of lemma 2.4, the corresponding ratio after half of its energy is extracted is at most $z_{\text{full}}(y/2)/z(y) \leq C$ for $y \geq 2$. Here z_{full} uses cutoff one. The upper and lower $y^{-3/2}$ estimates in that lemma prove this bound, so this normalization spends a bounded logarithmic price at each such site. These uses of positive energy require an allocation of disjoint shares, including multiplicities when carriers overlap. The same energy share cannot produce a gain in two factors of eq. 8.ii.

8.2. **Counting the standing labels.** We state the local quantitative inputs for the label estimate. They concern the exact zero-source representation at the stated stop. They are supplied by the regular and large-field steps on the actual carriers and reference laws. No bound on the sum over final supports is part of these inputs.

The cell side at level t is $M\check{R}_t$ in its lattice units. Write $\vartheta_t = \min_{i \leq t} \theta_i$. In this partition estimate specialize the cell exponent of eq. 3.v to a fixed $r \geq 1$. Use the actual rounded scale

$$\check{R}_t = L^{\lceil \log_L((\log \vartheta_t)^r) \rceil}.$$

The frame bound $\theta_i - \theta_t \geq -c_*$ for $i \leq t$ gives $\theta_t - c_* \leq \vartheta_t \leq \theta_t$. Comparability $x_t \asymp \theta_t$ on the real tube therefore gives $1 + \log x_t \leq C\check{R}_t$ above a fixed floor, while $\check{R}_t \leq L(\log \theta_t)^r$. If $\theta_t - \theta_{t+1} \leq B$, the running minimum drops by at most B in a step. Above a further fixed floor, $(\log \vartheta_t / \log(\vartheta_t - 2B))^r < L$. Consequently the rounded scale decreases by at most one power of L over two successive steps. The physical partitions, which are aligned, either agree or coarsen by L at each step, and two successive partitions cannot both agree with their predecessors.

The final cells form a finite graph by the adjacency used in the operation. Its maximum degree D_0 is fixed, and the number of cells is at most N_n. A label has a unique collection of connected standing supports in that graph. For a fixed collection, its remaining visible information is recorded separately under those supports. Internal records already summed in a retained mass need only be tracked insofar as the label distinguishes them. If labels distinguish their visible traces, the mass in eq. 8.iii is split according to those traces; summing the disjoint parts recovers the original mass. The record for a support may be bounded by a larger set of possibilities, with repetitions only if their counts are charged explicitly.

For each support C, root its record at a fixed final cell of C. The active addresses are level-cell pairs, including the final cells of C. They form a connected set by the parent and merger rules, with any necessary bounded collars included in the charge. At successive levels the partition either stays the same or coarsens by L, so a cell has at most L^4 children. A neighbor at the same level has bounded range in cell units. The address graph has fixed degree $D \geq 2$, also on a periodic quotient. For an active set $\mathcal{A}_C$ define its charged volume by

$$v_C = \sum_{(t,\mathfrak{c}) \in \mathcal{A}_C} (M\check{R}_t)^4 (1 + \log x_t).$$

This dominates both the number of its active addresses and $|C|$, the number of final cells in C. A fixed multiplicity of charges changes the cost constant. For a fixed active set the count of the other record choices, excluding normalized frame choices, is at most $\exp(bv_C)$ with a fixed b. In the actual local code this follows when a bounded alphabet records a bounded number of tests per site and bond, and choices of stage or failure type at level t have at most a fixed power of

$1 + \check{R}_t$ possibilities per active cell. Indeed $\log(1 + \check{R}_t) \le C(1 + \log x_t)$ for the chosen polynomial sizes in $\log x_t$. Multiplying the local counts gives the stated bound with the own-scale volume. Frame counts are integrated with their normalized laws, which have total mass at most one on any restriction. Every additional discrete choice must be included in this local code.

The second input is the local action and cost estimate. Separate the exponential remaining after removing $v_{n,\mathcal{S}}$ into the assigned energy factors, the retained integrands and the remaining regular, remainder and boundary exponent. Its positive part, together with the unused reference and vacuum costs, is bounded by

$$A_0(1 + \log x_n)N_n + A_1 \sum_C v_C. \tag{8.iv}$$

The first term is the contribution of the regular region in final units. The second is an own-scale estimate for terms meeting the recorded large-field regions. In particular the missing vacuum contributions are charged at their own levels: each omitted or reassigned numerical contribution has a charged carrier of bounded multiplicity and costs at most a constant times $1 + \log x_t$ per site at its level t. The same allocation applies to normalization costs of the reference integrals. Any part of the cost inside a retained operation is paid using eq. 8.iii at a reserve weight chosen before the inverse-coupling floor. When a record sees only the visible part of a drop-out, the remaining volume it sees is charged to that same operation at the raised weight. Its already integrated creations are not separate gains. Charges paid in this way can be removed from eq. 8.iv.

After these choices, the actual weighted retained masses and certified failed tests give a product of factors e^{-G_C}. For each fixed exterior record the analytic input consists of the local failure gains and mass bounds with the supremum in eq. 8.iii. In a component they supply gains $g(e)$ on its paying events, so $G_C = \sum_e g(e)$. Here e ranges over creations outside outermost retained carriers and the distinct retained marks with one certified, unused gain assigned to their visible successor. A creation outside those carriers supplies $g(e) \ge c_1 p_0(g_t^{\mathrm{ref}})^2$ in its own level t, or the reduced $c_1 p_0(g_t^{\mathrm{ref}})$ estimate after any preceding allocations. A mark supplies $g(e) \ge c_1 p_0(g_t^{\mathrm{ref}})$ either from a reserved part of its weighted mass or from a separately allocated last-test share. For a native drop-out, its inner weighted mass excludes the failed last test: that test supplies the gain for the renewed wait, while the inner mass pays its recorded loads. No gain is inferred merely from the presence of a mark, and each energy share is used once. We use $g_t^{\mathrm{ref}} = \theta_t^{-1/2}$ and $p_0(g_t^{\mathrm{ref}}) = A(\log \theta_t)^p$ for fixed $A, p > 0$. Every nonempty component has at least one paying event.

The geometric inputs for the genealogy charge are the following features of the actual hull and waiting rules. Each active component has a skeleton of occupied cells connected by bounded-range steps in the cell graph, and each charged cell is within a fixed cell distance of the skeleton. At the next level the occupied cells come from the parent images of predecessor skeletons and at most a bounded number of new occupied cells per paying event. Here a retained mark uses the visible skeleton after its own hidden records have been integrated. The seeds of an event enter at its own level or at a later level, as for a bare failure recorded at the first step. A cut leaf that needs a new gain uses an assigned last-test share outside the inner mass.

The fixed-distance property is compatible with a collar at every step. If an old collar has radius d in its cell units and the partition coarsens by q, its new radius is at most $d/q + A$ for a fixed A, including the parent-boundary ambiguity and the new collar. Over two levels the older radius contracts by L^{-1}. Its accumulated contribution is therefore bounded by $2A/(1 - L^{-1})$. Boxing only components of bounded diameter adds a fixed amount to A. The unique-address parent map is applied to the occupied cells, while the closed cell carriers and their collars enter the volume estimate.

A stretch with one predecessor and no new paying event evolves by the quiet rule. For a fixed skeleton-size threshold, once the skeleton has at most that many cells, at most $C\check{R}_t$ quiet levels may occur before another event, a join, the end of the genealogy, or a healing, if the stretch started

at level t. The constant may depend on the fixed threshold. This is the local waiting bound, with the size at the start of the stretch. We use it only on the non-erased genealogy. Each unerased predecessor has a unique successor containing its whole hull; components can merge but cannot split. Consequently the ancestry of one standing component is a rooted tree, directed toward that component. Every leaf has a paying event. The size $\check{R}_t$ is nonincreasing, and the partition has no two consecutive steps without coarsening.

The waiting bound uses an embedded light component and its required collar. If that light geometry has no embedded chart, its bounded cell diameter and bounded collars imply that some torus period in cell units is bounded. This period is divided by at least L every two further levels. Since it must stay an integer at least one, only a fixed number of further levels can be admitted. This is also covered by the waiting bound, by increasing its constant.

Lemma 8.2 (Own-scale charge of a standing genealogy)**.** Under the skeleton and waiting conditions above, the record volume is bounded by

$$v_C \le C_{\mathrm{geo}} \sum_e M^4 \check{R}^6_{t(e)}. \tag{8.v}$$

The sum uses the same paying events as the mass estimate and has no factor depending on the total depth. ■

Proof. Over s steps the parent partition coarsens by a factor $Q \ge L^{\lfloor s/2 \rfloor}$ in cell units. For a connected skeleton of m cells, walk twice along a spanning tree. The walk has total length at most Cm in the old cell units, with every step of bounded length. Divide it into at most $C(1 + m/Q)$ consecutive portions of length at most CQ. Each portion lies in a box of side $O(Q)$, hence meets only $O(1)$ parent cells. On a torus lift each portion before counting its parent cells; projection can only decrease the number. Thus the skeleton has at most $C_1(1 + m/Q)$ parent cells. Fix s so that $Q \ge 4C_1$. For $m \ge 4C_1$ the count is at most $m/2$; below this fixed size it is bounded by a constant.

Write o_j for the total number of occupied cells in the level-j members of the genealogy, l_j for the number of its members below the fixed size, and a_j for the number of events whose seeds enter at j. The parent rule, allowing a new occupied cell per event with fixed multiplicity, gives

$$o_j \le \tfrac{1}{2} o_{j-s} + C l_{j-s} + C \sum_{j-s<t\le j} a_t.$$

Use zero counts before the first level. For $r_j = M^4 \check{R}^5_j$, multiply by r_j and sum. Because r_j is nonincreasing, moving the first sum to the left gives

$$\sum_j r_j o_j \le C \sum_j r_j l_j + C \sum_t r_t a_t.$$

If an event pays at an earlier level than its insertion, monotonicity bounds its weight by the weight at its paying level. Bounded skeleton neighborhoods and $1 + \log x_j \le C\check{R}_j$ bound v_C by a constant times the left side.

Divide the genealogy tree into stretches by beginning a new stretch at a paying event, just after a keeping mark, or at a join. Along a stretch without an event the parent images cannot increase the number of occupied cells. Its light members therefore form a last segment, of length at most $C\check{R}_t$ for a start at t, by the quiet waiting bound. Its cost in $\sum_j r_j l_j$ is at most $C r_t \check{R}_t$. Charge an event start or a successor of a mark to its event or that preceding mark, respectively. To charge join starts, designate a first predecessor at each vertex with predecessors. At a join take a non-first predecessor, then follow first predecessors down to a leaf. This assigns distinct leaves to distinct joins. Indeed on the path up from the chosen leaf the first ascent from a non-first predecessor occurs at the specified join. A leaf is no later than the join and has a paying event. Monotonicity of $r_t \check{R}_t$ allows that earlier event to pay. Each event is used at most a fixed number of times. This bounds the right side of the last display by the right side of eq. 8.v. □

Different histories with one visible label can have different exterior creation sets. Extracting their largest remaining factor requires a common reference after all the preceding changes of variables and density factors. More precisely, suppose the hidden record h is integrated by a nonnegative kernel $K(dh \mid z)$ with $K(1 \mid z) \le 1$. If $H(h,z) \le M(z)$ on its support, then $\int H(h,z)K(dh \mid z) \le M(z)$. The bound $M(z)$ remains inside the z integral. To select a single worst record for the label, first extract a common nonnegative integrable envelope in z; the remaining outside factor, including its record-dependent numerical costs, must depend only on h. Its maximum is then a deterministic factor. The chosen record and visible trace must remain in the stated code count. The original stochastic operation alone does not provide this domination after a density is refactored. An alternative is to keep the actual disjoint record pieces and include their reference masses or multiplicities in the count before summing. One of these two warrants is part of the actual record input. Without it, m pieces of mass e^{-G} have total mass me^{-G}, which a maximum alone does not bound.

Lemma 8.3 (Rooted record count)**.** For each final support C, choose a fixed root in C. Suppose the record inputs just stated hold. There is a fixed c_{e} for which the sum of $e^{-c_{\mathrm{e}}v_C}$ over its possible outer records, with normalized frame laws, is at most one. ■

Proof. Order the neighbors of each vertex of the address graph. Associate to each connected active set of size m containing the root a deterministically chosen rooted spanning tree, then record the depth-first walk that traverses each tree edge twice. The visited set determines the active set, and the walk has length $2(m-1)$. Hence there are at most $D^{2(m-1)}$ choices of active set. The other choices for an active set use at most e^{bv_C} codes. Thus the sum is at most

$$\sum_{m\ge 1} D^{2(m-1)} e^{-(c_{\mathrm{e}}-b)m} \le 1$$

if $c_{\mathrm{e}} - b \ge 2\log D + \log 2$. The root level is fixed. Positions at earlier levels are part of the same walk, and do not cost a factor equal to the depth or the fine volume. □

Proposition 8.4 (Upper bound for the history sum)**.** Suppose the zero-source representation satisfies eq. 8.i, the integral and gain inputs of section 8.1, and the record, cost and genealogy inputs of section 8.2. Assume $\check{R}_t \le C(1+\log\theta_t)^r$, $p > 6r$, and $\theta_t \ge \theta_n - c_*$ for $t \le n$, with a fixed c_*. Then, with the inverse-coupling floor sufficiently large, uniformly in n, cutoff and admissible volume,

$$\widehat{Z}_n = e^{-E_{(n)}} Z_{K',\ell,\beta} \le \exp\{C(1+\log x_n)N_n\}. \tag{8.vi}$$

The same bound holds for the sum of absolute retained integrals with any inserted component weights already covered by the specified reserve. ■

Proof. Insert and remove the numerical factor $e^{v_{n,\mathcal{S}}}$ in the common normalization of each term of eq. 8.i. Bound the residual exponent by eq. 8.iv. The retained and failure factors are handled by lemma 8.1 with the reference product of the actual record, including the reserve for any inserted weight. The finite sums over disjoint pieces of an internal record can be taken inside the resulting majorants. We obtain an upper bound by a sum of products, with factor $e^{A_1 v_C - G_C}$ for each outer record of C, and the same factor $e^{A_0(1+\log x_n)N_n}$ for all labels.

Choose $\kappa > 2\log D_0$, using $D_0 \ge 2$ without loss of generality. Since $p > 6r$, the gain of each paying event exceeds $2(A_1+\kappa+c_{\mathrm{e}})C_{\mathrm{geo}}M^4\check{R}^6_{t(e)}$ when the common floor is large enough. The floor can also ensure $\log\theta_t \ge \frac{1}{2}\log\theta_n$ for $t \le n$. Thus one half of G_C pays $(A_1+\kappa+c_{\mathrm{e}})v_C$ by eq. 8.v. The other half leaves $s_n = c_{\mathrm{gain}}p_0(g_n^{\mathrm{ref}})$ for a fixed $c_{\mathrm{gain}} > 0$, because there is at least one paying event. It follows that

$$e^{A_1 v_C - G_C} \le e^{-s_n - \kappa|C|} e^{-c_{\mathrm{e}}v_C}.$$

The sum over records on a fixed C is at most $a(C) = e^{-s_n-\kappa|C|}$ by lemma 8.3. This step takes place before the sum over placements of the final supports.

Apply the same walk count in the final graph, now with degree D_0. The sum of $a(C)$ over all nonempty connected supports is at most

$$N_n e^{-s_n} \sum_{m\ge 1} D_0^{2(m-1)} e^{-\kappa m} \le C_0 N_n.$$

Finally, discard compatibility between distinct final supports in the sum over their collections. Every collection occurs at most once once its records are fixed. Its weight is nonnegative, so the sum is at most $\prod_C (1 + a(C)) \le \exp(\sum_C a(C))$. Combining the bounds proves eq. 8.vi. □

The reference to actual component integrals in this argument cannot be replaced by a pointwise bound on one polymer gas. If a representation is $Z = \int q(y)\Xi_y \, d\pi(y)$, where $q \ge 0$ and a real logarithm satisfies $|\log \Xi_y| \le cN_n$, then

$$e^{-cN_n} \int q \, d\pi \le Z \le e^{cN_n} \int q \, d\pi.$$

For an empty gas and q the indicator of a Haar set of measure $e^{-N_n^2}$, the logarithm vanishes and $Z = e^{-N_n^2}$. The lower bound below uses an explicit event with a controlled Haar measure.

8.3. **The upper bound with a specified label load.** The actual read ownership gives a more precise estimate when the remaining unmarked regular energy has an extensive bound in final units. We state this version with the scalar load in its own level units. On a standing label let Ω_j be the determining regions and let Λ_j^{a} be the small-field regions of the action sequence. Put $Z_j^{\mathrm{a}} = (\Lambda_j^{\mathrm{a}})^c$ and let Z_j^+ be its one-cell enlargement. Write $\#_j(D)$ for the number of level-j sites in D. Set $Y_0^{\mathrm{d}} = T_\ell \setminus \Omega_1$ and $Y_m^{\mathrm{d}} = \Omega_m \setminus \Omega_{m+1}$ for $1 \le m \le n$, with $\Omega_{n+1} = \varnothing$. Thus Y_0^{d} is the full outside layer in bare units. For $c \ge 0$, define

$$\mathscr{L}_c(\mathcal{S}) = c \sum_{m=0}^{n} \#_m(Y_m^{\mathrm{d}}) + \sum_{j=1}^{n} \{C_{\log} \log \theta_{j-1} + C_{\mathrm{loc}} + c\} \#_j(Z_j^+). \tag{8.vii}$$

At $n = 0$ use $Y_0^{\mathrm{d}} = T_\ell$. For each fixed c, the two price constants in eq. 8.vii are nonnegative. Fix them and the raised retained-integral reserve before the inverse-coupling floor. Use $\sigma_{\mathcal{S},c}$ for the literal absolute integral of this label after stripping its own factor $e^{v_{n,\mathcal{S}}}$ and inserting the factors $e^{c\,v(\omega)}$ at the outermost retained operations.

Proposition 8.5 (The loaded history estimate)**.** Use the actual product coordinates, reserve, and record inputs of proposition 8.4. Suppose the entire remaining final-site price after stripping the label vacuum is bounded by $C_{\mathrm{sf}} N_n$. This includes the unmarked energy and every regular, remainder, boundary or reference factor not already paid by the retained masses or assigned own-scale defect prices. The actual remaining Wilson factor uses the ordered profiles of eq. 2.vi, and its allocated shares leave nonnegative plaquette coefficients. The determining regions are nested with Ω_{m+1} aligned at level $m + 1$, and $\Lambda_j^{\mathrm{a}} \subset \Omega_j$. The frame has $1 < \theta_j$ and $\theta_{j-1} - \theta_j \le B$ with a fixed B. Use the disjoint final-cell partition from the standing geometry, in which every cell contains at least M^4 level-n sites. Assume the actual local read ownership. Every earlier layer address and enlarged defect address lies in the collar of its assigned action cell, or is assigned to the latest retained operation that remade that address. These assignments have bounded multiplicity in the original level of each address. For a remade portion of volume V_{in}, measured with its own logarithmic site weights, every retained record allowed by the visible label has $v(\omega) \ge V_{\mathrm{in}}$. Assume the local mass bounds hold at the correspondingly raised reserve, chosen before the inverse-coupling floor. Then

$$\sum_{\mathcal{S}} e^{\mathscr{L}_c(\mathcal{S})} \sigma_{\mathcal{S},c}(1) \le \exp\{(c + C_{\mathrm{sf}} + M^{-4}) N_n\}. \tag{8.viii}$$

If the price $\mathscr{L}_0$ bounds the positive omitted vacuum number $v_{n,\mathcal{S}} - E_{(n)}$, this also bounds $\widehat{Z}_n$ by the right side for $c = 0$. ■

Proof. The last layer contributes at most cN_n. For $m < n$,

$$\#_m(Y_m^{\mathrm{d}}) \le \#_m(\Omega_{m+1}^c) = L^4 \#_{m+1}(\Omega_{m+1}^c) \le L^4 \#_{m+1}(Z_{m+1}^{\mathrm{a}}).$$

The equality counts the level-m sites of each aligned level-$(m+1)$ block. The first inequality at $m = 0$ includes every bare site of the outside layer. Assign the enlarged Z_j^+ cells to neighboring original cells of Z_j^{a}, with bounded multiplicity. The one-step frame bound gives $\log \theta_{j-1} \le C(1 + \log \theta_j)$. It follows that the earlier portions of $\mathscr{L}_c$ are bounded by own-scale prices, with constants affine in c. Apply the read ownership to retain the assigned addresses when an operation remakes a determining layer. An ordinary address is paid from the genealogy's event reserve. A remade address is charged to the operation that owns it, in the original level units, even if several later terms read that operation.

For the latter cost first restrict to the compatible inner records $\mathcal{B}$ of that mark. If its assigned price is at most $A_c V_{\mathrm{in}}$, choose $c' \ge c + A_c$. On these records

$$e^{A_c V_{\mathrm{in}}} e^{cv(\omega)} \le e^{c' v(\omega)}.$$

Consequently the price times $\mathsf{M}_c(i, X; \mathcal{B})$ is bounded by $\mathsf{M}_{c'}(i, X; \mathcal{B})$, with the same restriction. The unrestricted raised mass is an upper bound for this restricted mass. Keeping the restriction when labels separate visible records preserves their disjoint sums. The older failures already integrated in the mark are not charged or used as a gain a second time. An unfinished live lineage has no completed mark mass; its actual failures and waiting costs are charged by the preceding genealogy estimate.

Apply the exterior-supremum estimate and the event prepayment of proposition 8.4 with these fixed, increased local costs. The surviving regular factor is at most $e^{(c+C_{\mathrm{sf}})N_n}$. By eq. 2.vi and the share allocation, the remaining real Wilson factor is at most one. The remaining Haar probability measure has mass at most one on the cut. Each final connected component C, after summation of its records, keeps the bound $e^{-s_n - \kappa|C|}$. Choose the fixed κ so that $\sum_{m \ge 1} D_0^{2(m-1)} e^{-\kappa m} \le 1$. There are at most N_n/M^4 possible root cells, since a final cell contains at least M^4 sites. The sum of these component weights is at most N_n/M^4. The product over distinct compatible components is bounded by its exponential, as in the proof of proposition 8.4. This gives eq. 8.viii. At $c = 0$ the absolute integral controls the vacuum-stripped label, so the final assertion follows by reinserting the common vacuum difference. □

8.4. **An event in the empty standing label.** For the empty label the determining sequence is constant and only the final bonds V remain. Use the actual small-field test χ_n and background $U_n(V)$ of this label. The supplied density formula is

$$e^{-E_{(n)}} d\tau_\emptyset(V) = \chi_n(V) \exp\{-x_n A(U_n(V)) + F_n(U_n(V))\} \prod_b dV_b. \tag{8.ix}$$

The equality of the numerical factor here with $E_{(n)}$ uses the same localization and normalizers as eq. 2.iv. On this label the exponent F_n contains the vacuum-subtracted regular and remainder terms, including those produced by healing. There is no standing boundary term.

We record the needed regularity inputs for this formula. For age $h = n - j$, the bound on the pointed, vacuum- and classically subtracted regular family is, per level-j base site,

$$C_E\{(1+h)L^{-b_{\mathrm{T}}h} + L^{-h}\}L^{-4h}, \qquad b_{\mathrm{T}} > 0.$$

It is evaluated on the constant determining sequence, where the partition functions used for pointing sum to one. The corresponding bound on the vacuum-subtracted remainder is $C_R(1+h)L^{-4h}\theta_j^{-q}$, with $q > 2$. These are rooted bounds on the actual test support. They include the localization tails on wrapping domains and any terms made by the large-field step. Assume $\theta_j \ge \theta_n + \lambda(n-j) - c_*$

for $\lambda > 0$ and $\theta_n \geq 1 + c_*$. There are $L^{4h}N_n$ base sites. Summing first at age h removes L^{-4h}; summing over ages uses the two convergent series

$$\sum_{h\geq 0}\{(1+h)L^{-b_{\mathrm{T}}h} + L^{-h}\}, \qquad \sum_{h\geq 0}(1+h)(1+\lambda h)^{-q}.$$

Consequently $|F_n| \leq C_I N_n$ on the stated support.

Lemma 8.6 (Admitting a product of link balls)**.** Suppose the actual terminal test uses the separated determining towers of the empty standing label, with all tested plaquettes in the interior test window. The actual determining datum on each tower, including the constant global sequence, is the centered face lift of the terminal bonds and its prescribed iterated averages, as described in the proof. This is an identity for the datum used by the operation. Assume the linear background regularity estimate on these towers and on the constant global sequence. Namely, a determining datum of plaquette radius d below a fixed ceiling gives plaquette defects at most $B_3 d L^{-2n}$ on the tested plaquettes and on the bare plaquettes of the global background, respectively. Suppose χ_n is exactly the product of these terminal plaquette tests on the admitted stock; any other terminal condition has already been verified to be one there. Each plaquette test is one below $\varepsilon_n L^{-2n}$, with $\inf_n \varepsilon_n\sqrt{\theta_n} > 0$ and $x_n \asymp \theta_n \geq 1$ uniformly. Then for some fixed $c_{\mathrm{sf}} \in (0, 1]$, on

$$\mathcal{B}_n = \{V : \|V_b - I\|_{\mathrm{op}} < c_{\mathrm{sf}} x_n^{-1/2} \text{ for every } b\}$$

one has $\chi_n = 1$ and

$$x_n A(U_n(V)) \leq 48 B_3^2 c_{\mathrm{sf}}^2 N_n. \tag{8.x}$$

At level zero the action estimate holds with $B_3 = 1$ without a background construction. ■

Proof. Put $s = c_{\mathrm{sf}} x_n^{-1/2}$. The product of the four unitary matrices around a final plaquette differs from the identity in operator norm by at most $4s$, by telescoping and unitarity. To compute the radius of the actual determining datum, lift V to finer links as follows. A link staying inside a top block is the identity. A link crossing a top block face is assigned its corresponding oriented top link. These prescriptions use the periodic centered blocks, or their common lift in a test window, and commute with reversal.

An elementary fine plaquette crossing faces in no coordinate has holonomy I. If it crosses faces in only one coordinate, the two crossing factors are inverses and the other two are identities. If it crosses in both coordinates, the four crossing factors form the corresponding top plaquette holonomy with its induced orientation. Thus its radius is at most $4s$. The one-step averaging of this link field is its lift at the coarser resolution. Indeed, at an intermediate scale the two endpoint contours are inside their own top blocks and contribute identities. Each joining segment on a coarse bond crosses either no top face or the same single face in the same orientation, so every transported candidate in the group mean has the same value. Centered nested blocks ensure this also for projected paths at a periodic face, counting the oriented path occurrences. Iteration gives the same radius on the averaged rows of each terminal tower. This is the radius of the actual datum when it is formed by the prescribed lift and averages.

The stated linear estimates now give defect at most $4B_3 s L^{-2n}$ on the tested stock and the global bare background. Choose c_{sf} small enough for $4s$ to be below the datum ceiling and $4B_3 s < \varepsilon_n$. The choices are uniform by the two comparisons in the statement. The test is then one. Earlier fluctuation tests are already incorporated in eq. 8.ix. For $U \in \mathrm{SU}(2)$ with class angle φ,

$$1 - \tfrac{1}{2}\operatorname{Tr} U = 1 - \cos\varphi = \tfrac{1}{2}\|U - I\|_{\mathrm{op}}^2.$$

There are $6N_0$ plaquettes in the bare Wilson sum and $N_0 = L^{4n}N_n$. The two powers of L^{-2n} cancel this site conversion and give $A(U_n(V)) \leq 48 B_3^2 s^2 N_n$. At $n = 0$ use the direct plaquette telescoping with the bare field V in the same calculation. □

Theorem E (Common-normalized partition bounds)**.** Under the history hypotheses of proposition 8.4 and the empty-label density, rooted estimates and regularity hypotheses of lemma 8.6, one has

$$e^{-(C+6\log x_n)N_n} \;\le\; e^{-E_{(n)}} Z_{K',\ell,\beta} \;\le\; e^{C(1+\log x_n)N_n}. \tag{8.xi}$$

At the terminal first passage $n = K$ with $h = g^{-2} \le x_K \le h + B_{\mathrm{s}}$ and $K' = K$, this gives

$$e^{-C_g \ell^4} \le e^{-E_{(K)}} Z_{K',\ell,\beta} \le e^{C_g \ell^4}.$$

The constants are independent of the bare cutoff and compatible volume. At earlier stops eq. 8.xi uses N_n and requires the same local inputs at that stop. ■□

Proof. For $0 < s \le 1$, an SU(2) matrix in the class-angle range $0 \le \varphi < s$ belongs to the operator-norm ball of radius s, since $\|U - I\|_{\mathrm{op}} = 2\sin(\varphi/2)$. The normalized Haar angle density is $(2/\pi)\sin^2\varphi\, d\varphi$. The inequality $\sin\varphi \ge 2\varphi/\pi$ for $0 \le \varphi \le 1$ therefore gives ball mass at least $8s^3/(3\pi^3)$. There are $4N_n$ independently integrated positive links.

By positivity, keep the empty label in eq. 8.i, then restrict its integral to $\mathcal{B}_n$. Use $|F_n| \le C_I N_n$ and eq. 8.x. With $c_H = 8/(3\pi^3)$ this gives the lower bound

$$e^{-E_{(n)}} Z_{K',\ell,\beta} \ge e^{-(C_I + 48 B_3^2 c_{\mathrm{sf}}^2)N_n} \big(c_H c_{\mathrm{sf}}^3 x_n^{-3/2}\big)^{4N_n}.$$

The upper bound is eq. 8.vi. At the first passage with $K' = K$, $N_K = \ell^4$ and the inverse coupling lies in a fixed interval depending on g. Absorb its endpoint into C_g. □

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